



Sharing information in the credit market: Contract-level evidence from U.S. firms[☆]



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ABSTRACT

We investigate the impact of lenders' information sharing on firms' performance in the credit market using rich contract-level data from a U.S. credit bureau. The staggered entry of lenders into the bureau offers a natural experiment to identify the effect of lenders' improved access to information. Consistent with the predictions of Padilla and Pagano (1997, 2000) and Pagano and Jappelli (1993), we find that information sharing reduces contract delinquencies and defaults, especially when firms are informationally opaque. The results also reveal that information sharing does not reduce the use of guarantees, that is, it may not loosen lending standards.

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1. Introduction

In recent decades, a broad consensus has formed that most credit market failures (such as credit rationing) are attributable to information asymmetries between lenders and borrowers. Although advances in information and communications technology are enhancing lenders' ability to acquire information on customers, the credit frictions stemming from information asymmetries are likely

to persist into the future (Petersen and Rajan, 2002). A response to these problems that is gaining ground is the development of institutional arrangements through which lenders share their proprietary information with one another. Examples of such arrangements are public credit registers and private credit bureaus, "informational intermediaries" that collect data on credit histories of firms and consumers and make them available to their members upon request. Credit registers and bureaus are numerous in developed countries (Jappelli and Pagano, 2002) and their presence is becoming pervasive in emerging and transition economies (Brown, Jappelli, and Pagano, 2009).¹

The benefits and costs of lenders' information sharing have been the object of increasing attention in the last 20 years or so. The conclusions reached by the theoretical literature suggest that lenders' information sharing generally improves the performance of borrowers, although

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¹ Conducting a survey of 43 countries, Jappelli and Pagano (2002) show that before 1950, less than 5% of the countries had a private credit bureau while in 2000, over 60% had one bureau at least.

the results are not always clear-cut. The seminal work of [Pagano and Jappelli \(1993\)](#) demonstrates that the information on the past behavior of borrowers made available by a credit bureau facilitates lenders' screening of credit applicants. Furthermore, lenders' information sharing can discipline borrowers ([Padilla and Pagano, 2000](#)) and deter lenders from holding up entrepreneurs ([Padilla and Pagano, 1997](#)), thus stimulating entrepreneurial effort. These mechanisms imply that the entry of a lender into a credit bureau should trigger an improvement in the repayment performance of its borrowers, such as a reduction in the delinquency and default rates. However, it is also acknowledged that lenders' information sharing may give lower-grade borrowers access to credit and that this could tend to increase delinquencies ([Jappelli and Pagano, 2006](#)). Clearly, establishing the impact of lenders' information sharing and gauging its magnitude can not only yield valuable insights into the functioning of the credit market, especially the consequences of information asymmetries, but also produce important normative implications. For instance, policy makers debate the optimal design of public credit registers and whether private credit bureaus should be subsidized.

The objective of this paper is to shed new light on the impact of lenders' information sharing on firms' performance in the credit market exploiting unique and unusually rich contract-level data from a U.S. credit bureau. The testing ground of our analysis consists of the equipment finance industry. At the end of 2000, PayNet Inc., a company engaged in providing predictive analytics for credit risk management, launched the "Payment Information Network," a credit bureau aimed at serving this industry. In the years that followed the inception of the bureau at the end of 2000, several major equipment finance companies underwent the process necessary to join the credit bureau. This involved an up-front investment in the technological infrastructure necessary to establish the interface with the bureau, negotiating an agreement with PayNet administrators, addressing legal issues arising from the disclosure of clients' confidential information, and developing a protocol for processing the information made available by the bureau. Importantly for our purposes, lenders completed these procedures at different speeds and, hence, joined the PayNet bureau on different dates: for instance, the 15 lenders investigated in this paper became members of the bureau in a period spanning from August 2001 to March 2004. Exploiting the staggered entry of lenders into the credit bureau as a natural experiment, we can isolate the contribution of the entry of a lender on the performance of its credit contracts from macroeconomic trends and from changes in the conditions of the industry.

Our data set consists of a random sample of the PayNet Payment Information Network containing about 28,000 loans and leases granted by 15 equipment finance companies to almost 4,000 businesses. The data set includes information on the performance of equipment finance contracts originated both before and after lenders' entry into the credit bureau. This information comprises several measures of contract delinquency and default, such as the number of days payments were late, the frequency of

moderate and serious delinquencies, and the occurrence of legal actions and debt collections. The data also inform us on the date of entry of each lender into the bureau, thus allowing us to exploit the staggered entry of lenders as a natural experiment. Finally, the sample offers details on the characteristics of firms and contracts. Most borrowers are small businesses, which is ideally suited for our purposes. In fact, small businesses are supposedly informationally opaque because they are not monitored by rating agencies or the financial press ([Petersen and Rajan, 1994](#)). Therefore, the benefits from joining a credit bureau should be particularly strong for lenders financing such firms.

After controlling for lender, firm, and contract attributes and for macroeconomic effects, we find that the entry of lenders into the credit bureau reduced the incidence of contract delinquencies and defaults. For the average contract, a lender that joined the bureau experienced a drop from 29.65 to 23.31 in the maximum number of days and a drop from 6.58 to 5.76 in the average number of days the borrower was late on his payments. Furthermore, for the average contract, a lender that joined the bureau experienced a reduction from 9.5% to 7.1% in the probability of a serious delinquency (90 or more days past due) and an equally sizable drop in the probability of a major event of default such as a debt collection or a legal action. As we shall elaborate, these results are consistent with the predictions of [Pagano and Jappelli \(1993\)](#) and [Padilla and Pagano \(1997, 2000\)](#) that lenders' information sharing can improve borrowers' repayment performance by allowing better screening of borrowers and mitigating borrowers' and lenders' moral hazard. However, interestingly, the analysis also reveals that lender entry into the bureau tended to raise the probability of bankruptcy. This result may suggest that delinquencies have an option value in that they give borrowers some flexibility in making regular payments. By reducing delinquencies, lender entry into the bureau could have forced a few borrowers to eventually resort to bankruptcy to prevent the repossession of assets.

In further tests, we take a closer look at the mechanisms through which lender entry into the bureau affected borrowers' performance. We uncover evidence that its impact was more pronounced for small and young businesses. This result is in line with expectations: sharing information on informationally opaque (e.g., small and young) firms is particularly helpful to lenders. The results also reveal that the impact of information sharing operated both along the intensive margin (an improvement in the performance of existing borrowers) and along the extensive margin (an improvement in the quality of the pool of borrowers). An additional question that our data help address is whether information pooling changes lending practices. We document that the entry of lenders into the credit bureau did not reduce the use of guarantees (indeed, it increased it for low rating firms), thus suggesting that lenders' information sharing does not necessarily loosen lending standards.

The plan of the paper is as follows. In [Section 2](#), we relate the analysis to the prior literature. [Section 3](#) describes the equipment finance industry and the credit

bureau under scrutiny. In Section 4, we draw testable hypotheses from the extant theoretical studies, devoting particular attention to characteristics of the equipment finance industry (e.g., the widespread availability of collateral) that can influence the impact of lenders' information sharing. Section 5 details the empirical methodology and the data. In Sections 6 and 7, we present the main results. Section 8 contains additional tests that explore the impact of information sharing on the form of financial arrangements. Section 9 concludes.

2. Prior literature

This paper relates to two strands of literature. The first strand investigates theoretically the role of information sharing in the credit market and includes papers such as Pagano and Jappelli (1993), Padilla and Pagano (1997, 2000), and Bennardo, Pagano, and Piccolo (2010).² We shall discuss extensively these theoretical models and their predictions. The second stream of studies investigates the consequences of information exchange empirically. Most of these studies rely on aggregate, cross-country data and explore whether in a country the presence of public credit registers and private credit bureaus is associated with the development of the credit market and the tightness of financing constraints. Jappelli and Pagano (2002) consider a panel of 43 countries and show that in countries with more credit registers, default rates are lower and bank credit is more abundant. Examining a set of 129 countries, Djankov, McLiesh, and Shleifer (2007) uncover evidence that especially in poor countries the presence of credit registers positively correlates with the ratio of private sector credit to gross domestic product (GDP) and that this ratio rises following the introduction of registers. Love and Mylenko (2003) obtain that the presence of credit registers and bureaus in a country lowers the perceived incidence of financing constraints, with this effect being differentially stronger for small and young firms. Galindo and Miller (2001) obtain analogous results exploring the impact of credit registers and bureaus in over 30 countries. While insightful, the analyses that employ aggregate data in cross-country empirical settings are exposed to measurement and endogeneity issues. For example, in such settings it is problematic to control for country-specific institutions and laws that affect credit market performance.

Probably because of a dearth of data, very few studies provide microeconomic evidence on the implications of lenders' information sharing. Kalberg and Udell (2003) describe the functioning of a private business information exchange and examine whether the credit performance score compiled by this information exchange helps predict the probability of firms' failure. Luoto, McIntosh, and Wydick (2007) and de Janvry, McIntosh, and Sadoulet (2010) focus on the provision of microfinance to solidarity groups and poor households in a developing country.

Using branch-level data from a microfinance institution (MFI), Luoto, McIntosh, and Wydick (2007) estimate that the introduction of a credit bureau in Guatemala improved the average performance of loans in the branches. Moreover, de Janvry, McIntosh, and Sadoulet (2010) find that the introduction of the bureau also affected the process of formation of solidarity groups, leading to higher turnover and the ejection of bad performing borrowers from the solidarity groups. Brown and Zehnder (2007) offer experimental evidence that a credit register motivates borrowers to repay loans. They implement an experimental credit market and show that if borrowers and lenders interact only once, the credit market will collapse, possibly because lenders fear that borrowers will default. The introduction of a credit register in this context raises repayment rates and the amount of credit granted. The contribution of our study to the empirical literature is unique in its kind in that we can exploit the natural experiment generated by the staggered entry of lenders into a credit bureau in conjunction with rich contract-level data on U.S. businesses.

3. Institutional background and data

In this section, we highlight the main features of the equipment finance industry and of the credit bureau investigated in the empirical analysis.

3.1. The equipment finance industry

Investment in equipment and software accounts for a large share of firms' total investment. In 2007, for example, it accounted for about 48% of the total gross private domestic investment and 67% of the nonresidential fixed investment in the United States (Bureau of Economic Analysis, 2012). According to the Equipment Leasing and Finance Association (ELFA, the trade association that serves as the advocate for the equipment finance industry to the U.S. federal government), equipment finance companies fund over 50% of the annual investment in equipment and software made by U.S. businesses. Moreover, according to ELFA, roughly 80% of U.S. firms use equipment finance.³ ELFA tracks the pattern of activity of the equipment finance industry by conducting a survey of prominent equipment finance companies.⁴ Between 1998 and 2007, on average, the net portfolio of the surveyed companies increased by 4.5% per year. Originations of

³ ELFA reports that its members finance the acquisition of all types of equipment and software such as manufacturing and mining machinery and equipment, vessels and containers, construction and off-road equipment, medical technology and equipment, commercial and corporate aircraft, trucks and transportation equipment, business, retail, and office equipment, IT equipment and software. In general, the transportation (rail, truck, marine, and aircraft) industries are the most likely to seek external finance for their investment in equipment. For example, in 2006, businesses in truck transportation externally financed almost 80% of their investment.

⁴ This information is extracted from the ELFA Monthly Leasing and Finance Index (MLFI-25) for the 2005–2007 period and from the Quarterly Performance Indicators Report (PIR) for the 1998–2004 period. These surveys track the performance of the same companies over time so they provide a fairly reliable trend analysis.

² A broad theoretical literature explores the role of financial intermediaries in screening and monitoring borrowers (see, e.g., Diamond, 1984; and for a review, Gorton and Winton, 2003).

new contracts fluctuated between a 25% drop in 2000 and a 22.8% rise in 2004.

Equipment finance companies include commercial banks, independent finance and leasing companies, captive finance companies, diversified financial services companies, as well as broker/packagegers and investment banks. Depending on the amount due, equipment finance contracts are classified into small-ticket, when less than a quarter of a million dollars, middle-ticket, when between a quarter of a million and five million, and big-ticket when over five million dollars. Traditionally, the lease has been the most common type of contract. In a lease, the equipment finance company (lessor) owns the equipment and rents it to the customer firm (lessee), which can have the option to buy the equipment during or after the term of the lease. Recently, however, other contracts have risen in importance and, according to ELFA, they now amount to over 50% of the total. These contract varieties include, among others, loans, in which the customer firm acquires, uses, and eventually owns the equipment after investing in a down payment; conditional sales, which are essentially leases in which, after the term, the lessor transfers the title to the lessee; and financing leases, which are leases in which the customer has control of the asset for a large portion of its useful life, and also has many of the benefits and costs of ownership, such as maintenance, taxes, and insurance. Loans are by far the most used among these alternative contractual arrangements.

The equipment finance industry is characterized by a widespread availability of collateral. Leases are intrinsically backed by the equipment that is rented to the lessee while loans can also be collateralized by the equipment that is acquired by the borrower. The reader could then wonder whether the availability of collateral reduces the need for information sharing in this industry. As we argue below, we expect the availability of collateral to possibly attenuate but not eliminate the benefits of lenders' information sharing.

3.2. The credit bureau

Our source of data is the PayNet "Payment Information Network" database. PayNet is a major company that provides credit risk management tools for business lending and, as stated on its official Web site, has the objective "of reducing operating expenses, increasing approval rates, controlling delinquencies, and reducing both actual write-offs and reserving requirements." PayNet provides to commercial lenders and credit card issuers credit ratings on commercial loans, leases, and other financial obligations for thousands of privately held small and medium-sized businesses. At the end of 2000, in partnership with the Equipment Leasing and Finance Association, PayNet launched a credit bureau to serve the equipment finance industry (the "Payment Information Network"). ELFA stresses that this bureau was the first of its kind in the industry and thus represented a major breakthrough for lenders' ability to share information. For example, in a report published in 2001 in the magazine of ELFA (ELT, 2001), an expert defined the PayNet bureau "a

breakthrough source of more relevant lessee credit history."⁵ Absent credit bureaus, financiers have to either acquire costly information or rely on informal contacts among local managers and loan officers (Jappelli and Pagano, 2006). According to ELFA, before the inception of the "Payment Information Network," equipment finance companies interested in pooling information could rely on very few small credit bureaus that provided only trade information on short-term debt obligations (e.g., the payment of utility bills). These limited information sharing arrangements were of little help for lenders' monitoring and screening of borrowers.⁶ The fact that the bureau was perceived as a new, major source of information is also revealed indirectly by the fact that equipment finance companies were willing to sustain sizeable costs for joining it, including an investment in technological infrastructure.

In the years following the inception of the PayNet bureau, several major equipment finance companies took the steps necessary to join it. This involved an up-front investment in a technological facility, solving the legal issues associated with the disclosure of confidential information on customers, reaching an agreement with PayNet administrators, and developing a protocol to process the credit reports made available by the bureau. In the documents accompanying the introduction of the bureau, the administrators of PayNet stressed that the up-front technological setup was more problematic and time consuming for some equipment finance companies than for others because "a certain percentage of the industry's players have custom systems, which requires a unique (technological) solution." Indeed, a major problem for some institutions was "getting IT resources to do the extractions (of credit reports)." Crucially for the objectives of this paper, because of these different technological and legal hurdles, lenders completed the procedures necessary for becoming members of the bureau at different speeds and ended up joining the bureau on different dates. We thus have strong reasons to believe that lenders' different dates of entry reflect the different technological and legal problems they faced in joining the bureau and not factors endogenous to the performance of borrowers or lenders' expectations regarding the benefits of the bureau. A further element that corroborates this view is that ELFA exerted strong pressure on equipment financiers to enter the bureau as soon as possible to enhance the comprehensiveness of the bureau. Therefore, it is very likely that the time needed for the technological, infrastructural, and legal setup was the binding constraint in determining lenders' entry dates.

The PayNet bureau is a "closed" network that operates on a principle of reciprocity: a lender gains access to information in the bureau only after supplying its own information. Specifically, when they join the bureau, lenders release information on their customers to the bureau. In exchange for this, they gain the right to retrieve reports

⁵ ELFA and PayNet are fully independent from each other, so we do not expect that the opinions of ELFA about the credit bureau were affected by PayNet.

⁶ This problem was probably stronger for some types of financiers, such as leasing companies, that are less frequently affiliated with credit bureaus. However, our data do not provide details on the types of creditors.

on credit applicants from the bureau, subject to the payment of very small fees.⁷ A credit report compiled by the bureau includes “black” or “negative” information on the applicant’s past credit delinquencies and defaults and “white” or “positive” information on some of the applicant’s characteristics, such as its PayNet credit rating, number of employees, level of indebtedness, and patterns of repayments.⁸ Although in the United States loans and leases for business purposes are not subject to the same privacy regulation as consumer loans, lenders systematically provide privacy notifications to businesses. Therefore, we expect that the borrowing firms in our sample were fully aware of the possible use of information on their repayment history by a credit bureau. A thornier issue is whether the firms in the sample quickly became aware of lenders’ affiliation to the bureau after entry occurred. As it will become clear shortly, firms’ awareness of lenders’ entry into the bureau matters for some of the mechanisms through which information sharing can impact firms’ repayment performance.

The PayNet bureau has a number of appealing features. First, unlike private credit bureaus directly owned by a consortium of lenders, it is independently managed by a third party. According to several studies, this should minimize the risk that the management of the bureau is negatively influenced by conflicts of interest among its members (see, e.g., *International Finance Corporation, 2006*). A second characteristic of the bureau is its reliability. The accuracy of its data is ensured by rigorous checking mechanisms and by strict norms that punish lenders with the exclusion from the use of the database in the event of misreporting or falsification of information.⁹ A third feature of the bureau is its comprehensive coverage. Like other private credit bureaus and unlike public credit registers, the PayNet bureau provides information on the single leases and loans issued to an applicant, rather than consolidated information on an applicant’s credit history. Moreover, it provides information on different categories of contracts. Because of lack of robust data on small businesses, prior to the inception of the PayNet bureau commercial lenders in the equipment finance industry had to rely on short-term payment histories to make medium- and long-term financing decisions. For instance, the few existing credit bureaus only reported to a lender how an applicant had paid his light bills, utility bills, or overnight express bills, all very short-term obligations which are poor predictors of repayment performance for equipment loans, leases, and other such contracts. The PayNet bureau

covers contracts of all maturities, allowing equipment finance companies to make medium- and long-term lending decisions based on comparable credit.

4. Theoretical predictions

This paper focuses on the impact of lenders’ information sharing on borrowers’ repayment performance. In addition, we shall briefly investigate its impact on contractual terms. Based on the theoretical literature, we can identify three channels whereby information sharing can affect lending outcomes in the equipment finance industry.

(i) Adverse selection. The access to information provided by other financiers allows a lender to evaluate the credit worthiness of potential customers more effectively. In *Pagano and Jappelli (1993)*, a lender possesses more information about firms that operate in its local market than about firms in other markets. When a bank receives credit applications from businesses located in the area of a second bank, the affiliation of both banks to a credit bureau will allow the first bank to learn which applicants are less likely to default. However, *Jappelli and Pagano (2006)* also argue that *ex ante* it is ambiguous whether, after the formation of a bureau, credit will be granted only to higher quality applicants. In fact, lenders’ information sharing could facilitate the access to credit of lower-grade borrowers previously excluded from the credit market.

(ii) Lenders’ moral hazard. A second mechanism through which lenders’ information sharing can affect borrowers’ repayment performance is the hold-up channel. In *Padilla and Pagano (1997)*, each bank possesses private information on the quality of an entrepreneur. As in *Rajan (1992)*, after extending a loan to an entrepreneur a bank can exploit its private information on his quality and threaten to withhold credit to extract rents from him (hold up). Anticipating that the returns of his effort will be (partially) appropriated by the bank, the entrepreneur has then a reduced incentive to exert effort *ex ante*. In turn, this worsens his repayment performance. Banks can tackle this problem by committing *ex ante* to sharing with one another their proprietary information about entrepreneurs’ quality. Expecting that this information pooling will promote competition among lenders, entrepreneurs will be reassured that no hold up will be possible and will step up their effort, lowering delinquency and default rates.

(iii) Entrepreneurs’ moral hazard. A third channel through which a credit bureau can affect lending outcomes is by imposing discipline on borrowers. In *Padilla and Pagano (2000)*, lenders’ information sharing induces entrepreneurs to exert effort because they “perform for a broader audience,” that is, if they do not fulfill their contractual obligations, their misconduct will be disclosed to more lenders.¹⁰ However, *Padilla and Pagano (2000)* also underscore that this effect weakens if lenders pool information on borrowers’ characteristics in addition to

⁷ PayNet reports are delivered in two ways, PayNet On-Line and PayNet Direct. Members enter the PayNet On-Line site through a login/password-protected section. The member queries for the applicant and chooses from a menu of products. The report is then delivered to the screen for review and printing. In case multiple companies are identified from the inquiry, they will all be displayed to the member. PayNet data can also be sent directly to the credit-scoring process through PayNet Direct, a central processing unit (CPU)-to-CPU link.

⁸ Some credit bureaus only provide information about arrears and defaults while others inform on characteristics of borrowers such as assets and liabilities, guarantees, history of repayments, and debt maturity structure.

⁹ PayNet has a dispute resolution process that ensures compliance with state and federal credit-reporting laws. For a theoretical analysis of the incentives of a lender to behave opportunistically and provide false information to a credit bureau, see, e.g., *Semenova (2008)*.

¹⁰ *Diamond (1991)* puts forth a model in which firms establish a good reputation as borrowers so as to obtain less expensive credit in the future.

information on delinquencies and defaults. In this case, a high quality entrepreneur knows that his high quality will be disclosed to lenders regardless of whether his credit history is good or bad.¹¹

To recapitulate, as argued by Jappelli and Pagano (2006), the theoretical literature predicts that the effect of lenders' information sharing on the performance of contracts is ambiguous a priori. The sign of this effect depends on whether lenders' improved ability to screen borrowers and the reduction in lenders' and borrowers' moral hazard prevail or not over the easier access to credit of low-grade borrowers.

As anticipated, the widespread availability of collateral in the equipment finance industry could attenuate adverse selection and moral hazard problems, reducing the need for lenders' information sharing. Extant theories suggest that collateral can mitigate adverse selection by facilitating the screening of applicants who are more likely to default (see, e.g., Besanko and Thakor, 1987). Moreover, lenders can discourage entrepreneurs' moral hazard by threatening to liquidate collateral. Yet, scholars and practitioners agree that collateral is far from entirely solving adverse selection and moral hazard problems, so that lenders' information sharing is likely to retain an important role. First of all, collateral cannot cover all the losses associated with firms' delinquency and default. Second, in some cases, it can even exacerbate rather than mitigate entrepreneurs' and lenders' moral hazard.

Let us consider the first argument. Collateral can be illiquid so that the liquidation costs a lender has to sustain can make its recovery value significantly lower than the value reported in the balance sheet or at the moment of contract origination. There is the problem of limited marketability: collateral can be a piece of equipment designed specifically for a firm and without a large secondary market. There is also the problem of the uncertainty in the market value of pledged equipment (see, e.g., Bennardo, Pagano, and Piccolo, 2010 for a model in which there is default because the value of collateral is uncertain). A third problem is that the value of collateral can drop over time because of depreciation and obsolescence and, in particular, some collateral can end up having a life shorter than the life of the contract. The problems of limited marketability, price instability, and perishability of collateral are so important that practitioners agree that collateral and information about credit histories should be viewed as complements in improving lending outcomes.¹²

The second reason for which lenders' information sharing retains its importance is that collateral itself can generate distortions in the behavior of borrowers and lenders. Berger and Udell (1990) show that managers can care little about collateralized assets, underinvesting in

their maintenance, or engage in perk consumption from such assets. Collateral can also induce moral hazard on the part of lenders. Manove, Padilla, and Pagano (2001) demonstrate that collateral can make bank officers lazy in screening applicants. Expecting this effect on loan officers, banks may want to share information useful for better screening. Furthermore, Berger and Udell (1990) and Carey, Post, and Sharpe (1998) argue that financial institutions that specialize in collateralized lending tend to lend to riskier borrowers.

Overall, do we expect the availability of collateral to be more beneficial in ameliorating adverse selection or moral hazard problems? Berger, Frame, and Ioannidou (2011) find that collateral tends to mitigate moral hazard problems more than adverse selection ones. Their findings might suggest that lenders' information sharing retains a more beneficial role in mitigating adverse selection problems. However, as the authors argue in their review of the literature, the empirical evidence on these issues is still very limited so no clear conclusion can be drawn.

5. Evidence from contract performance

In the first part of the empirical analysis, we carry out tests that collapse the repayment history of each contract into one observation, such as the average or maximum delinquency rate over the life of the contract or the occurrence of a major event of delinquency or default (e.g., a serious delinquency, debt collection, or legal action). This approach has the cost of neglecting information on the internal dynamics of contracts, complicating the way we can control for macroeconomic conditions. However, it has the important benefit that we do not need to exclude early contracts from our tests. In fact, before the first quarter of 2000, we lack period-by-period information on the payments made to lenders and only have information on the overall performance of a contract. In this section, we first discuss the empirical methodology. After that, we detail the measurement of the variables and present sample summary statistics. The Appendix contains a description of the variables used in the analysis.

5.1. The empirical model

The baseline empirical model we estimate is

$$\begin{aligned} bad_{ijs} = & c + \alpha Information_{js} + \sum_j \beta_j D_j^L + \sum_i \gamma_i D_i^F + \sum_t \delta_t D_t^S \\ & + \sum_t \varepsilon_t D_t^C + \zeta C_{ijs} + \eta M_{ijs} + u_{ijs}. \end{aligned} \quad (1)$$

The dependent variable bad_{ijs} is a measure of the (poor) performance of the contract extended to firm i by lender j on date s . We consider various measures of the poor performance of a contract that span from moderate delinquencies, to major events, such as serious delinquencies, legal actions, and debt collections.

Consider first moderate delinquencies. Our measures include the natural logarithm of (one plus) the average number of days a payment was past due during the contract (*average DPD*) and the natural log of (one plus) the maximum number of days past due during the

¹¹ Bennardo, Pagano, and Piccolo (2010) predict that lenders' information exchange can mitigate delinquencies by reducing the risk of firms' overindebtedness. This aspect appears to be of limited importance in an industry where collateral is often available.

¹² Experts in the equipment finance industry also mentioned to us the additional problem that sometimes the borrower steals the equipment pledged as collateral (e.g., cars, trucks).

contract (*maximum DPD*). One could also argue that the cost of delinquencies may not be proportional to the payment delay (for instance, two 15-day delinquencies could imply a lower cost than a 30-day delinquency). Therefore, as a further indicator we consider the log of (one plus) the number of times during the life of the contract that a payment was 30 or more days past due. To verify whether the effects of information sharing on moderate delinquencies operate only on the intensive margin, we complement these three measures with a (0,1) dichotomous variable that takes the value of one if at least once during the contract a payment was late (*delinquency*).

Delinquencies carry several costs for lenders. They imply interest revenues lower than expected because of delayed payments. They also slow down the turnover of lenders' portfolios, affecting lenders' cash flows and liquidity. Furthermore, delinquencies entail costs due to more extensive monitoring and analysis of loan and lease portfolios, sometimes visits to borrowers, and actions for pursuing delinquent borrowers. The resources and time a financier invests in controlling delinquencies also mean lower ability to expand services. Practitioners often stress that delinquencies also cause loss of credibility and lower staff morale. For all these reasons, they are used extensively to assess the quality of lenders' portfolios (for an analysis of the equipment finance industry, see Murtagh, 2005) and to construct indicators of the credit performance of businesses (see, e.g., Kalberg and Udell, 2003, and references therein).

In addition to measures of moderate delinquency, our data set includes information on major events, such as serious delinquencies (90 or more days past due), legal actions, debt collections, and bankruptcies. Therefore, we also employ this information to construct the dependent variable in (1). It is not straightforward how to group major events. A first problem is that there are various, not easily comparable events. We can broadly classify them into: (i) events such that the lender still considered the loan or lease recoverable, such as serious delinquencies; (ii) events such that the loan or lease was considered at least partially unrecoverable, such as legal actions, debt collections, forced extensions, and corresponding accounting treatments (e.g., write-offs); (iii) episodes in which the borrower filed for bankruptcy. A second problem associated with major events is that in the data there is some overlap among the various events of default, so that treating any of them in isolation encounters some measurement issues. For example, in the data set, a firm that first was involved in a legal action and subsequently experienced a debt collection is reported as "debt collection." Finally, we have to consider the low frequency of these major events. Keeping these caveats in mind, we believe that the most reasonable approach is to group the major events into four indicators based on the classification in (i)–(iii). The first two indicators capture serious delinquencies: the log of (one plus) the number of serious delinquencies (*number of times over 90 DPD*) and a dummy variable that takes the value of one if a serious delinquency occurred, zero otherwise (*serious delinquency*). The third indicator for major events is a

comprehensive indicator variable that takes the value of one if the firm incurred any of the major events of default listed in (ii), zero otherwise (*major event of default*). Finally, the fourth indicator is a dummy variable that takes the value of one if the firm went bankrupt, zero otherwise. Bankruptcy is conceptually different from the actions in (ii), since those actions are initiated by the creditor while bankruptcy is initiated by the debtor. Moreover, Section 365(d)(4) of the Bankruptcy Code leaves to firms the freedom to reject or assume leases. These observations suggest that it makes sense to analyze bankruptcy separately.

On the right-hand side of (1), we include a constant c , a variety of controls, and our key variable of interest, a time-varying treatment variable (*information*) that takes the value of one if the contract started no sooner than the date lender j joined the bureau, zero otherwise. Control variables include lender dummies D_j^L , reflecting differences in delinquency or default rates due to lender-specific time-invariant characteristics, and firm dummies D_i^F , capturing time-invariant firm attributes (for comparison, in some tables we also display regressions without firm dummies).¹³ We also insert time dummies to control for macroeconomic conditions. Since a contract spans various periods and we are treating each contract as a single observation, we let time effects be determined by the start date and the close date of the contract. Thus, we include a dummy D_t^S that takes the value of one if the start date S of the contract is in period t and a dummy D_t^C that takes the value of one if the close date C of the contract is in period t . Also, in estimating model (1) we consider a period to be a year, but we verify that the results carry through when a period is defined as a quarter. Finally, we include two contract-level control variables. The first is a dummy C_{ijs} for the type of deal that takes the value of one if the contract is a loan or a related contract, zero otherwise. Loans differ from leases in various dimensions mentioned earlier, including a different treatment under Chapter 11. The second contract-level control is contract maturity (M_{ijs}). This could be a determinant of our measures of contract performance, especially in the case of the number of late payments (*number of times over 30 DPD* and *over 90 DPD*) because this number may tend to increase with the passage of time.¹⁴ Due to their potential endogeneity to lending outcomes, we initially exclude additional contract-level controls (such as the size of the contract) from the model. In fact, unobserved factors could drive both contract performance and the choice of contract terms. In robustness tests, we augment the specification with further contract attributes.

When using dichotomous dependent variables (e.g., the dummies for the occurrence of major events of delinquency or default), we estimate the model in (1)

¹³ Standard errors heteroskedasticity-robust and clustered at the firm level.

¹⁴ In the literature it is often claimed that the maturity of a contract closely matches the nature of the equipment purchased or leased. In turn, the nature of the equipment is likely to reflect the technological needs of the borrowing firm so that overall we expect endogeneity problems to be mild when inserting contract maturity in the regressions.

both by ordinary least squares (OLS) and by probit. Furthermore, our nondichotomous dependent variables (e.g., the number of serious delinquencies) constitute a corner solution response, taking the value of zero for a nontrivial share of the population and being otherwise distributed over positive values. To account for this, when using such variables we estimate the model in (1) both by OLS and by Tobit. As explained by Wooldridge (2008), the OLS method provides a good approximation to the conditional expectation of the dependent variable, especially for values of the covariates close to the mean values. However, OLS estimates also present some drawbacks. First, in the case of dichotomous dependent variables, the predicted probability can fall outside the zero-one range and in the case of corner solution responses, the predicted value of the dependent variable can be negative. Second, the partial effect of any explanatory variable expressed in levels is constrained to be constant. Complementing the OLS regressions with probit and Tobit estimates allows to assuage these concerns.

5.2. Measurement and sample properties

Our data set consists of a random sample of the portfolios of 15 equipment finance companies that gradually joined the PayNet bureau after its inception. The data set comprises detailed information on firm and especially contract performance and characteristics. Table 1, Panels A to E, reports summary statistics. The total number of borrowing firms in the sample is 3,815. The firms are located in all U.S. states, the District of Columbia, and Puerto Rico, and span the majority of two-digit Standard Industrial Classification (SIC) codes. Of the 2,133 firms that report their SIC code, about 15% operate in construction, slightly more than 11% in manufacturing, 17% in transportation, 18% in wholesale or retail trade, and 28% in services. While we do not have demographic information for all 3,815 firms, for the 1,757 businesses for which the number of employees in 2007 is available, the average number of employees in that year was approximately 131 and the median was eight. Revenues, which are also reported as of 2007 and are available for 1,613 firms, amounted to \$114 million, on average, with a median of \$750,000. We compute the average age of firms based on the start date of contracts. That is, for a given firm, we take the average of the firm's age as of the start dates of its contracts. For the 2,140 firms for which information on age is available, we find that age was 16 years, on average, with a median of 12. Computing the average firm rating in an analogous fashion, we find that the average rating was 44.8 (out of 100), with a median of 42.5. If, instead, we consider the weighted average of firm ratings (that is, we compute the average rating across contracts), we obtain that the average rating was 54 and the median was 55. These ratings are in the ballpark of those of all PayNet members: we obtained documentation that reports that in 2001, for instance, the average rating across all new contract originations was 56.

To better grasp their magnitude, we compared the demographic statistics for the firms in our sample with those for the pooled 1998 and 1993 waves of the National Survey of Small Business Finances (NSSBF) conducted by the U.S. Board of Governors of the Federal Reserve System

and the Small Business Administration. On average, the businesses in the pooled NSSBF waves are 15 years old (with a median of 12) and have 30 employees (with a median of six). Thus, the businesses in our sample are slightly larger than those in the NSSBF, although they are still small or medium-sized. As noted, the small and medium size of the firms in the database is appropriate for our purposes. In fact, public information on small businesses is not readily available because they are not covered by bond rating agencies or the financial press (Petersen and Rajan, 1994). Therefore, the affiliation to a credit bureau is likely to convey to a lender information that could not be obtained otherwise.

At the contract level, the data set provides information on 28,623 deals originated by 15 lenders (Table 1, Panel B reports the percentages of contracts originated by the various lenders). Many lenders hold a portfolio of contracts diversified across industrial sectors, with three out of the 15 lenders having more than half of their portfolio concentrated in one sector (two lenders in transportation, another in construction). Panels C to E display summary statistics for contract start dates, maturities, sizes, and for the measures of contract performance. The oldest contracts in the sample started as early as the fourth quarter of 1986, while the latest started in the second quarter of 2007. However, start dates are not uniformly distributed over the sample period, as the large majority of them (28,320) are after 1994. Regarding the information available to lenders, about 59% of the contracts in our sample started after the lender joined the PayNet bureau. In almost 90% of cases, we know whether there is equipment related to the contract. In practically all of these contracts, there is indeed equipment that is being leased or purchased, with the most common equipment types being trucks, construction equipment, and copy machinery. With regard to contract maturity, the average is 44 months and the median 47. Contract size, as given by the total amount to be paid back to the lender, is on average equal to \$102,633, and the median is \$43,813. Thus, most contracts are small. In fact, about 93% of contracts are referred to as “small-ticket,” that is, under \$250,000, and all of the remaining contracts, except for 33, are “medium-ticket” since they are below \$5 million. Contracts have a guarantor in approximately 35% of cases, while in 55% of deals there is no guarantor (in the remaining 10% of cases we lack this information).

6. The estimation results

In this section, we start by presenting the core empirical results. Afterward, we enrich the baseline specification and carry out extensions. In estimating the model in (1), we generally restrict attention to contracts that were closed as of the second quarter of 2007 (the last quarter in our sample). In fact, we need to account for the possibility that the impact of information is greater at some point during the life span of a contract and, if that is the case, including contracts that are still open at the end of the sample could bias our results. We also exclude the oldest contracts, meant as those started before January 1995. In fact, while contracts in the data set started as early as 1986, 99% of the start dates are between 1995 and 2007. This could raise concerns about the

Table 1

Summary statistics.

This table reports firm-level (Panel A) and contract-level (Panel C) summary statistics for the sample as well as the percentages of contracts by lender (Panel B). The table also displays summary statistics for contracts started before (Panel D) and after (Panel E) lenders' entry into the PayNet bureau. The total number of firms is 3,815, but firm revenue (in dollars), number of employees, age (in years), and rating are available for 1,613, 1,757, 2,140, and 3,486 firms, respectively. Revenue and number of employees are reported as of 2007. Rating and age are reported as of the start date of the contract. The total number of contracts is 28,623. Contract maturity is expressed in months, contract size is measured in dollars and reflects the total amount the firm has to repay, not the amount borrowed. Average DPD and maximum DPD are, respectively, the average and the maximum number of days past due during the contract; No. of times DPD over 30 and over 90 are the number of times the contract was 30 or more days past due and 90 or more days past due (for all these delinquency measures, we report summary statistics in levels, not logs). Delinquency is a dummy that takes the value of one if during the contract at least once a payment was past due, zero otherwise; serious delinquency is a dummy that takes the value of one if during the contract at least once a payment was 90 or more days past due, zero otherwise. Major event of default is an indicator variable taking the value of one if during the contract a major event of default (e.g., a legal action or a debt collection) occurred, zero otherwise. Bankruptcy is an indicator variable taking the value of one if during the contract the firm went bankrupt, zero otherwise. Guarantor is a dummy equal to one if the contract has a guarantor, zero otherwise.

Panel A: Firm-level summary statistics

	Revenue (\$)	No. of employees	Age (years)	Rating
No. of observations	1,613	1,757	2,140	3,486
Mean	114,000,000	131	16.09	44.80
Standard deviation	778,000,000	1,635	15.65	23.96
25th Percentile	238,000	2	5.60	25.00
Median	750,000	8	11.70	42.50
75th Percentile	3,000,000	26	21.10	62.50

Panel B: Number of contracts by lender

Lender	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
Number	1,603	1,231	3,234	716	1,116	2,118	716	4,843	1,603	4,694	2,662	494	1,403	2,118	71	28,623
Percentage	5.6	4.3	11.3	2.5	3.9	7.4	2.5	16.9	5.6	16.4	9.3	1.7	4.9	7.4	0.2	100.0

Panel C: Contract-level summary statistics

	Start date	Contract maturity (months)	Contract size (\$)	Average DPD (days)	Maximum DPD (days)	No. of times DPD Over 30	No. of times DPD Over 90
No. of observations	28,622	28,622	28,622	17,216	21,588	21,588	21,588
Mean	9/13/2002	43.726	102,633	6.195	29.532	2.156	0.447
Standard deviation	1000 days	16.840	404,012	21.065	51.783	5.283	2.371
25th Percentile	12/14/2000	36	16,728	0	0	0	0
Median	12/20/2002	47	43,813	0	10	0	0
75th Percentile	9/27/2004	60	92,504	3	32	2	0

	Delinquency	Serious delinquency	Major event of default	Bankruptcy	Guarantor
No. of observations	21,588	21,588	21,588	21,588	24,868
Mean	0.601	0.096	0.023	0.015	0.346
Standard deviation	0.490	0.295	0.189	0.122	0.476

Type of contract

	Conditional sale	Loan	Lease purchase	Rental lease	Revolver Account	True lease	Unknown	Total
Number	4,649	12,357	813	848	459	8,974	522	28,622
Percentage	16.2	43.2	2.8	3.0	1.6	31.4	1.8	100

Equipment type

	Number	Contract size (\$)	Contract maturity	Percentage
				Guarantor
				Lease
Agricultural	1,005	78,795	43.66	37
Automobile	242	30,359	44.53	34
Buses and motor coaches	658	127,589	54.61	35
Construction	5,885	111,271	41.13	36
Computer	1,535	108,131	37.39	32
Copy and fax	2,350	21,475	47.18	34
Forklift	639	76,630	38.29	31
Medium/light duty truck	1,668	66,608	47.51	30

Table 1 (continued)

Manufacturing	1,000	193,980	43.08	30	47
Office	170	94,330	45.14	39	40
Retail	1,011	62,177	42.31	38	35
Telecommunications	342	91,139	40.94	38	32
Truck	7,633	109,926	47.30	36	12
Vending and restaurant	503	26,726	48.24	44	17
Waste and refuse handling	78	916,463	53.26	23	46
Unknown	3,416	77,939	42.57	32	57
None	487	73,540	36.97	30	1
Total	28,622	102,633	43.73	35	31

Payment frequency				
	Number		Percentage	
Monthly	25,340		88.5	
Quarterly	173		0.6	
Semiannual	2,717		9.5	
Annual	369		1.3	
Other	23		0.1	
Total	28,622		100.0	

Panel D: Contract-level summary statistics before entry							
	Start date	Contract maturity (months)	Contract size (\$)	Average DPD (days)	Maximum DPD (days)	No. of times DPD	
						Over 30	Over 90
No. of observations	11,599	11,599	11,599	8,817	11,449	11,449	11,449
Mean	1/28/2000	44.736	107,482	7.807	34.307	2.724	0.571
Standard deviation	731 days	16.21	368,123	24.772	57.661	6.265	2.893
25th Percentile	2/26/1999	36	16,869	0	0	0	0
Median	7/26/2000	48	46,276	0	19	0	0
75th Percentile	6/6/2001	60	93,051	5	39	3	0
	Delinquency	Serious delinquency	Major event of default	Bankruptcy	Guarantor		
No. of observations	11,449	11,449	11,449	11,449	10,356		
Mean	0.668	0.106	0.024	0.017	0.315		
Standard deviation	0.471	0.308	0.141	0.129	0.465		

Panel E: Contract-level summary statistics after entry							
	Start date	Contract maturity (months)	Contract size (\$)	Average DPD (days)	Maximum DPD (days)	No. of times DPD	
						Over 30	Over 90
No. of observations	17,024	17,024	17,024	8,399	10,139	10,139	10,139
Mean	6/27/2004	43.037	99,329	4.504	24.139	0.463	0.308
Standard deviation	515 days	17.222	426,719	16.118	43.596	1.755	1.575
25th Percentile	5/25/2003	36	16,626	0	0	0	0
Median	5/25/2004	42	41,829	0	2	0	0
75th Percentile	8/12/2005	60	92,186	1	31	1	0
	Delinquency	Serious delinquency	Major event of default	Bankruptcy	Guarantor		
No. of observations	10,139	10,139	10,139	10,139	14,512		
Mean	0.525	0.085	0.022	0.013	0.367		
Standard deviation	0.499	0.279	0.135	0.114	0.482		

representativeness of the earliest contracts in the sample. In any case, at the end of this section, we will verify that the results are robust to including such contracts. Once

we remove old contracts and contracts still open at the end of the sample, we are left with 16,675 deals for which we know the average number of days past due and 20,904

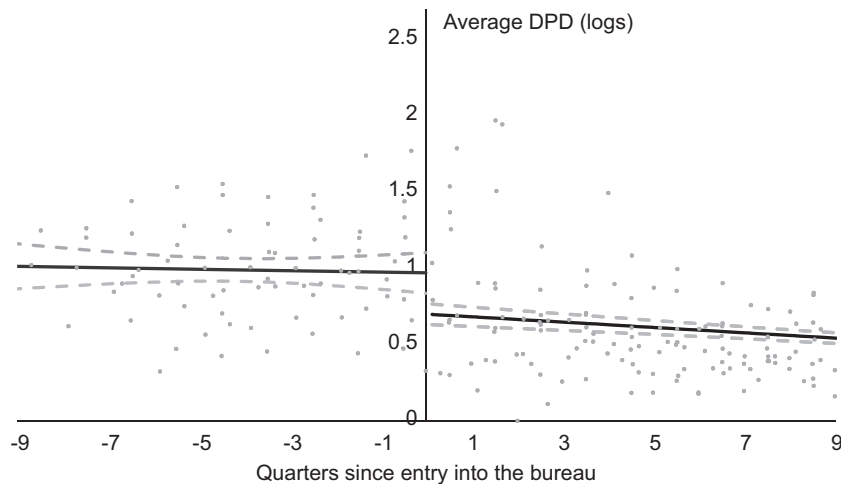


Fig. 1. Change in borrowers' performance after lenders' entry into the bureau. This figure reports average delinquencies (number of days payments are past due) in a quarter and by lender prior to and after lenders' entry into the bureau.

deals for which we have information on the other measures of contract performance.¹⁵ In both of these samples, approximately half of the contracts started after lenders joined the bureau.

6.1. Exogeneity tests

The inception of the PayNet bureau offers a clean natural experiment so that we do not expect endogeneity problems to affect our results. However, to further assuage concerns, we perform some exogeneity tests. One hypothesis we can verify is whether the timing of entry of lenders into the bureau was correlated with transitory shocks to the performance of their portfolios around the dates of their entry, that is, there is an “Ashenfelter dip” (Ashenfelter, 1978) in our data. Finding no such correlation would suggest that lenders did not join the bureau on different dates because they were motivated by a different evolution of the performance of their portfolios. Fig. 1 displays the average delinquencies in lenders' portfolios (number of days payments are past due, in logs) prior to and following lenders' entry into the bureau. The figure also reports a trend line together with the bounds of a 95% confidence interval.¹⁶ There is a clear discontinuity when the lenders join the bureau while there appears to be no pattern in the performance of lenders' portfolios prior to their entry into the bureau.

¹⁵ For 20,904 contracts, the data set reports the maximum number of days past due during the contract as well as information on the other indicators of contract delinquency and default. For 16,675 contracts, the data set also reports the number of days past due in each quarter (for the remaining contracts, we do not have information on days past due in all quarters). Using this information, we can compute the average number of days past due during the contract.

¹⁶ The trend lines are obtained by regressing the average delinquencies in a lender's portfolio on time before and after entry into the bureau. Observe that in the figure there are data points falling in between quarters. In fact, the entry of some lenders occurred in between quarters while the data for contract delinquencies are reported at the beginning of each quarter.

Analogous figures for other measures of average contract performance (e.g., the share of contracts in a lender's portfolio 30 or more days past due in a quarter) essentially lead to the same conclusion. The insights offered by Fig. 1 are confirmed by two tests. The first test is displayed in Table 2. Using our sample, we regress measures of average contract performance for a lender in a given quarter on lender dummies, on quarter dummies, and on dummies capturing the quarter of entry into the bureau and the quarters before and after entry. In all the regressions the omitted dummy refers to the quarter before entry while the other dummies referring to the previous pre-entry quarters, to the quarter of entry, and to the post-entry quarters are included. We construct the measures of average performance of a lender's portfolio in a quarter using the indicators of contract delinquency and default described in Section 5.1. In particular, we consider the average delinquencies (number of days past due) in the quarter, with the average taken across all the contracts of the lender. We also consider the share of contracts in the lender's portfolio that reach the maximum payment delay in the quarter, that are 30 or more days and 90 or more days past due in the quarter, as well as the share of contracts that end in the quarter and for which there is a major event of default or a bankruptcy. For all these measures, the coefficient estimates on the pre-entry dummies support the hypothesis of independence between lenders' time of entry into the bureau and the performance of their portfolios in the quarters before entry (see Table 2). Moreover, the coefficient estimates on the post-entry dummies reveal an improvement in the average contract performance (e.g., a reduction in delinquencies) in the quarters after entry (consistent with the results we will obtain in the rest of the paper, there is instead some evidence of an increase in the share of bankruptcies).¹⁷ The second exogeneity test we carry out

¹⁷ The dummy referring to the quarter of entry (i.e., the quarter Q0) turns out to be statistically insignificant in all the regressions. As Table 2

Table 2

Exogeneity tests.

This table reports coefficient estimates and associated standard errors (in parentheses) for tests of exogeneity of the entry of lenders into the PayNet bureau. Average days past due (DPD) in portfolio is the average number of days the contracts of a lender were delinquent in a quarter. Maximum DPD in portfolio is the share of contracts of a lender that reached the maximum payment delay in the quarter. Share of contracts over 30 DPD and share of contracts over 90 DPD are respectively the share of contracts of a lender that were 30 or more days late and 90 or more days late in the quarter. Share of contracts with major event of default and share of contracts with bankruptcy are, respectively, the shares of contracts that ended in the quarter and for which there was a major event of default (e.g., a legal action or a debt collection) and a bankruptcy. For sample details, see Table 1. The regressions are estimated using the sample of the analysis and include lender dummies, quarter dummies, and dummies capturing the quarter of entry into the bureau (quarter “Q0”) and the quarters before and after entry into the bureau. In all the regressions the omitted dummy refers to the quarter before entry, that is, the quarter “–Q1,” while the dummies for the previous pre-entry quarters, the quarter of entry, and the post-entry quarters are included. The table reports estimated coefficients for the pre-entry dummies referring to quarters “–Q8” to “–Q2” and the post-entry dummies referring to quarters “+Q1” to “+Q8.” All the regressions are estimated by ordinary least squares (OLS). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. For all the regressions, the table also reports the adjusted R-squared.

	Average DPD in portfolio	Maximum DPD in portfolio	Share of contracts over 30 DPD	Share of contracts over 90 DPD	Share of contracts with major event of default	Share of contracts with bankruptcy
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)
–Q8	9.367** (4.409)	0.206 (0.225)	0.171 (0.107)	0.051 (0.050)	–0.096 (0.137)	–0.087 (0.065)
–Q7	7.537* (3.720)	0.189 (0.193)	0.132 (0.091)	0.047 (0.044)	–0.088 (0.118)	–0.055 (0.048)
–Q6	6.313* (3.416)	0.199 (0.164)	0.122 (0.078)	0.032 (0.028)	–0.089 (0.105)	–0.044 (0.040)
–Q5	3.856 (3.098)	0.125 (0.136)	0.082 (0.064)	0.030 (0.023)	–0.069 (0.087)	–0.051 (0.032)
–Q4	2.422 (2.137)	0.097 (0.107)	0.074 (0.051)	0.021 (0.018)	–0.055 (0.067)	–0.039 (0.030)
–Q3	1.912 (2.034)	0.068 (0.079)	0.045 (0.037)	0.021 (0.014)	–0.046 (0.051)	–0.033 (0.022)
–Q2	1.213 (1.654)	0.044 (0.052)	0.018 (0.024)	0.013 (0.009)	–0.032 (0.033)	–0.018 (0.016)
+Q1	–2.189 (1.386)	–0.068 (0.049)	–0.036 (0.023)	–0.007 (0.009)	0.021 (0.022)	0.017 (0.013)
+Q2	–2.569** (1.148)	–0.126* (0.076)	–0.063* (0.036)	–0.010 (0.013)	0.036 (0.040)	0.023 (0.018)
+Q3	–6.618*** (1.773)	–0.179* (0.104)	–0.123** (0.049)	–0.018 (0.018)	0.049 (0.057)	0.035 (0.025)
+Q4	–5.706*** (1.838)	–0.170 (0.134)	–0.141** (0.063)	–0.023 (0.023)	0.052 (0.074)	0.026 (0.033)
+Q5	–7.354*** (2.221)	–0.214 (0.162)	–0.158** (0.076)	–0.027* (0.015)	0.071 (0.090)	0.056 (0.037)
+Q6	–9.026*** (2.950)	–0.251 (0.190)	–0.190** (0.090)	–0.036* (0.021)	0.084 (0.108)	0.068 (0.050)
+Q7	–10.021*** (3.455)	–0.273 (0.219)	–0.213** (0.103)	–0.035 (0.038)	0.114 (0.126)	0.110* (0.056)
+Q8	–12.699*** (4.129)	–0.314 (0.248)	–0.248** (0.117)	–0.045 (0.043)	0.110 (0.143)	0.075 (0.060)
Adj R-sq	0.644	0.412	0.462	0.308	0.456	0.318

consists of a Spearman test, which computes the degree of association between the ranking of lenders by entry date and the ranking of lenders by average delinquencies in the quarters before and after entry. We consider up to eight quarters before and after entry. The results of the

Spearman test (not tabulated to conserve space) confirm the hypothesis of independence between lenders' time of entry into the bureau and the performance of their portfolios in the quarters around entry. Collectively, these results further support the idea that lenders' time of entry was driven by exogenous factors.

6.2. The baseline results

In Table 3, we report the estimates of model (1) in which we use measures of moderate delinquency to construct the dependent variable; in Table 4, we report the regressions focusing on major events of delinquency, default, and bankruptcy. In Table 3, columns 1–3, our dependent variable is a *delinquency* dummy which takes

(footnote continued)

shows, after entry there appears to be an improvement in all the measures of contract performance, with the exception of the share of major events of default in lenders' portfolios. However, the variable for defaults (share of contracts that end in the quarter and for which there is a major event of default) tends to capture contracts at the end of their life. In the rest of the paper, we will find that lender entry into the bureau primarily affects contracts started after entry, so the variable for defaults in Table 2 captures little of this effect.

Table 3

Information sharing and contract performance—moderate delinquencies.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on moderate delinquencies. Delinquency is an indicator variable taking the value of one if there was at least one late payment during the contract, zero otherwise. Average days past due (DPD) and maximum days past due (MPD) are, respectively, the average and the maximum number of days payments were past due during the contract (both in logs). Number of times over 30 DPD is a variable capturing the (log) number of times that payments on the contract were 30 or more days late. Information is a dummy variable that takes the value of one if the contract began after lender entry into the PayNet bureau, zero otherwise. The regressions include controls for contract maturity (in months) and contract type (a loan dummy variable which takes the value of one if the contract is a loan, zero otherwise). The regressions also include lender dummies and time dummies controlling for both the start date and the close date of the contract. Sample details are in Table 1. For delinquency, the regressions are estimated by OLS (linear probability) with firm dummies, and probit, with and without firm dummies (heteroskedasticity-robust standard errors, clustered at the firm level). For average DPD, maximum DPD, and number of times over 30 DPD, the regressions are estimated by ordinary least squares (OLS) and by Tobit, with and without firm dummies (robust, clustered standard errors). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. For all the regressions, the table also reports the adjusted *R*-squared (pseudo *R*-squared if probit or Tobit).

	Delinquency (yes/no)			Average DPD			Maximum DPD			Number of times over 30 DPD		
	OLS (1)	Probit (2)	Probit (3)	OLS (4)	Tobit (5)	Tobit (6)	OLS (7)	Tobit (8)	Tobit (9)	OLS (10)	Tobit (11)	Tobit (12)
Information	−0.067*** (0.016)	−0.321*** (0.068)	−0.249*** (0.079)	−0.114*** (0.033)	−0.314*** (0.069)	−0.766*** (0.155)	−0.232*** (0.054)	−0.405*** (0.078)	−0.474*** (0.173)	−0.044* (0.024)	−0.152*** (0.052)	−0.087 (0.135)
Contract maturity	0.001*** (0.0002)	0.004*** (0.001)	0.007*** (0.001)	0.003*** (0.001)	0.005*** (0.001)	0.023*** (0.003)	0.004*** (0.001)	0.002*** (0.001)	0.017*** (0.003)	0.002*** (0.0004)	0.003*** (0.001)	0.011*** (0.002)
Loan dummy	−0.043*** (0.017)	−0.173*** (0.063)	−0.358*** (0.079)	−0.115*** (0.039)	−0.271*** (0.079)	−0.789*** (0.168)	−0.135*** (0.060)	−0.250*** (0.087)	−0.800*** (0.186)	−0.030 (0.026)	−0.127*** (0.054)	−0.385*** (0.120)
Firm dummies	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No
No. of observations	20,904	15,469	20,901	16,675	16,675	16,675	20,904	20,904	20,904	20,904	20,904	20,904
Adj <i>R</i> -sq (pseudo if probit/Tobit)	0.367	0.363	0.142	0.729	0.388	0.628	0.616	0.248	0.051	0.675	0.395	0.052

the value of one if during the contract the borrower missed at least one scheduled payment, zero otherwise. In column 1 we present the results of a linear probability model, in column 2 those of a probit model. Since in the probit model the inclusion of firm dummies entails a loss of observations (a number of firm dummies perfectly predict the outcome), in column 3 we also report the results of a probit model without firm dummies. The *information* dummy negatively affects the probability of delinquency, suggesting that the entry of a lender into the bureau improved the performance of its contracts. In columns 4–12 of Table 4, we display the results of the OLS and Tobit estimates of model (1) in which we use the (log of the) average number of days past due, the (log of the) maximum number of days past due, and the (log of the) number of times payments were 30 or more days past due as dependent variables. Since the OLS model is likely to give a good estimate of the average effects (Wooldridge, 2008), and to avoid cluttering the tables, we omit the partial effects computed from the Tobit model (details available from the authors).

We obtain a consistently negative effect of the treatment variable on the measures of moderate delinquency. When delinquency is measured by the average number of days past due, the estimated coefficient on *information* (significant at the 1% level) implies that for the average contract lender entry into the bureau makes the average days past due fall from 6.58 to 5.76 (see column 4 in Table 3). When delinquency is measured by the number of times 30 or more days past due, the estimated coefficient on *information* implies that for the average contract lender entry into the bureau makes this number drop from 2.16 to 2.02 (column 10). However, the most sizeable effect appears to be that of *information* on the maximum number of days past due: the estimated coefficient implies that for the average contract lender entry into the bureau makes this number drop from 29.65 to 23.31 (column 7). A possible interpretation for this figure is that, rather than reducing his delinquencies uniformly over a contract, a borrower was especially careful to avoid spikes in his delinquencies. Indeed, some documents of ELFA mention that various lenders have policies that make the approval of a new credit application more difficult if the applicant has ever been delinquent for more than a certain number of days (see, e.g., ELT, 2002). Regarding the control variables, it shall be noted that the maturity of the contract appears to have a positive effect on the measures of moderate delinquency. Moreover, some lender dummies are statistically significant, suggesting that some financiers systematically experience lower delinquency rates than others.

To gauge the pecuniary benefit of the bureau due to the estimated reduction in moderate delinquencies, we performed some back-of-the-envelope calculations. To compute the benefit in terms of lower forgone interest earnings, we obtained the average annual bank prime loan rate from the first quarter of 1995 to the second quarter of 2007 from the Federal Reserve Economic Data (FRED) database of the Federal Reserve Bank of Saint Louis, 2012 and then divided this rate by 365 to obtain the approximate average daily interest rate. Next, we

Table 4

Information sharing and contract performance—major events of delinquency, default, and bankruptcy.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on major events of delinquency, default, and bankruptcy. Number of times over 90 DPD is a variable capturing the (log) number of times that payments on the contract were 90 or more days late. Serious delinquency is an indicator variable taking the value of one if there was at least one payment 90 or more days late during the contract, zero otherwise. Major event of default is an indicator variable taking the value of one if during the contract a major event of default (e.g., a legal action or a debt collection) occurred, zero otherwise. Bankruptcy is an indicator variable taking the value of one if during the contract the firm went bankrupt, zero otherwise. Information is a dummy variable that takes the value of one if the contract began after lender entry into the PayNet bureau, zero otherwise. The regressions control for contract maturity (in months) and contract type (a Loan dummy variable which takes the value of one if the contract is a loan, zero otherwise). The regressions also include lender dummies and time dummies controlling for both the start date and the close date of the contract. Sample details are in Table 1. For the number of times over 90 DPD, the regressions are estimated by ordinary least squares (OLS) and by Tobit, both with and without firm dummies (heteroskedasticity-robust standard errors, clustered at the firm level). For the other measures of contract performance, the regressions are estimated by OLS (linear probability) with firm dummies, and by probit, with and without firm dummies (robust, clustered standard errors). For Major event of default and for Bankruptcy, the probit estimates are not reported because the probits did not converge. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. For all the regressions, the table also reports the adjusted R-squared (pseudo R-squared if probit/Tobit).

	Number of times over 90 DPD			Serious delinquency (yes/no)			Major event of default	Bankruptcy
	OLS (1)	Tobit (2)	Tobit (3)	OLS (4)	Probit (5)	Probit (6)	OLS (7)	OLS (8)
Information	−0.047*** (0.017)	−0.524*** (0.017)	−0.626*** (0.059)	−0.024*** (0.009)	−0.444*** (0.115)	−0.243* (0.148)	−0.005** (0.002)	0.008** (0.003)
Contract maturity	0.001*** (0.0003)	0.009*** (0.0002)	0.025*** (0.001)	0.001*** (0.0002)	0.008*** (0.002)	0.010*** (0.002)	0.0001** (0.00005)	0.0004*** (0.0001)
Loan dummy	−0.007 (0.014)	0.002 (0.018)	−0.034 (0.057)	−0.003 (0.009)	−0.010 (0.109)	−0.016 (0.077)	0.003** (0.001)	0.015*** (0.004)
Firm dummies	Yes	Yes	No	Yes	Yes	No	Yes	Yes
No. of observations	20,904	20,904	20,904	20,904	6,250	20,901	20,904	20,904
Adj R-sq (pseudo R-sq if probit/Tobit)	0.341	0.486	0.049	0.572	0.331	0.066	0.609	0.617

multiplied this daily rate by the estimated reduction in the average number of days past due and by the average value of contracts, obtaining an average per-contract pecuniary benefit of slightly more than \$16 associated with lower lost interests. We performed a similar calculation to gauge the pecuniary benefit of the bureau due to lower fixed costs for monitoring delinquencies and pursuing delinquent borrowers. These costs include letters to borrowers, hours spent to analyze the delinquency and to visit the borrower, etc. Clearly, there is a large variety in such costs. Based on reports and fees published by some debt recovery agencies, we set at \$100 the total fixed cost for monitoring a delinquent borrower, sending letters, etc., when the borrower is more than 30 days in arrear. Using this value, the pecuniary benefit implied by our estimates in terms of lower fixed monitoring costs amounts to about \$14 per contract.¹⁸ Overall, the total pecuniary benefit due to the reduction of delinquencies is then about \$30 per contract, excluding further benefits more difficult to gauge, such as higher portfolio liquidity, and hence better management of the loan portfolio, and higher staff morale. On the cost side, an estimate of the costs for joining the bureau is not possible. However, considering the large number of loans and leases granted by a major equipment financier over the years, by themselves (that is, without accounting for any pecuniary

benefit due to a reduction of defaults) such pecuniary benefits could have been enough to justify the entry of a lender into the bureau.

In Table 4, we turn to estimate the impact of lender entry into the bureau on major events of delinquency, default, and bankruptcy. The first two measures consist of the (log) number of serious delinquencies (90 or more days past due) and of a dummy variable that takes the value of one if the contract had at least one serious delinquency. As an alternative, comprehensive indicator, we define a dummy variable that takes the value of one if there was a major event of default (e.g., a legal action or a debt collection). As discussed earlier, we exclude “bankruptcy” from this measure. The estimates are consistent with those for moderate delinquencies in that they point to a negative impact of the treatment variable on the incidence of major events of delinquency or default. Using the measures of serious delinquency, we estimate a drop from 0.441 to 0.375 in the number of serious delinquencies for the average contract and a reduction from 9.5% to 7.1% in the probability of a serious delinquency (see, respectively, columns 1 and 4 in Table 4). We also estimate a reduction from 2.3% to 1.8% in the probability of a major event of default (see column 7).

Interestingly, the estimates in column 8 reveal that, in contrast with the results discussed so far, the impact of *information* on bankruptcy is significantly positive.¹⁹ We propose two possible interpretations for this finding. First,

¹⁸ Details of the computations are as follows. The average annual bank prime loan rate was 7.16% in the 1995Q1–2007Q2 period, the average value of a contract is \$100,199, and the estimated reduction in the average number of days past due is 0.82. In turn, the estimated drop in the number of delinquencies over 30 days is 0.14 and the fixed cost for monitoring a delinquency is set equal to \$100.

¹⁹ A strand of literature stresses the complexity of the link between delinquencies on one side and terminations of contracts and bankruptcies on the other (see, e.g., Ambrose and Capone, 1996, for a discussion).

delinquencies may have an option value in that they give borrowers some flexibility in making regular payments (see, e.g., Ambrose and Capone, 1996; Ambrose, Buttimer, and Capone, 1997; Phillips and VanderHoff, 2004; Capozza and Thomson, 2006, and references therein). The entry of a lender into the bureau could limit the use of delinquencies as a “pressure relief valve” and force some borrowers to eventually resort to bankruptcy to gain time and prevent the repossession of assets. A second interpretation builds on the observation that defaulters belong to two different categories, those who choose to default (sometimes labeled “ruthless defaulters”) and those who are genuinely unable to repay lenders because of unexpected events (“trigger-event defaulters”). The entry of a lender into the bureau could induce a selection in the composition of borrowers, so that more ruthless borrowers could be screened away and relatively more trigger-event borrowers be left in lenders’ portfolios. As argued by Jappelli and Pagano (2006) and as discussed earlier in our review of the theoretical predictions, one might conjecture that the introduction of the bureau induced lenders to fund more risky but honest trigger-event borrowers. Trigger-event borrowers are exactly those who can benefit more from bankruptcy for stopping asset repossessions.

In the regressions, we dropped the oldest contracts in the sample based on their limited representativeness. As a robustness check, we reinserted these contracts and found that the results are robust to this exercise. For example, for the average number of days past due, the estimated coefficient of the treatment variable is significant at the 1% level and for the average contract it implies a drop from 6.67 to 5.81 in this number. A robustness check in a similar vein consists of reinserting in the sample the contracts still open in the second quarter of 2007 (the last quarter of the sample). The results are robust to this exercise too.

6.3. Further contract attributes

In the baseline specification, we have been parsimonious in enlisting contract characteristics as control variables. One rationale for this approach is that such characteristics could be determined by – and, hence, be endogenous to – the expected performance of the contract. Thus, inserting them may be a source of bias. Nevertheless, it is useful to verify the robustness of the results to the inclusion of further contract attributes. The data set provides information on the size of the contracts (in dollars), the kind of equipment purchased or leased, the presence of a guarantor and its type, and the payment frequency. The results obtained adding controls for these characteristics to the model in (1) are in Table 5.²⁰ The finding that the treatment variable negatively affects

moderate delinquencies and major events of delinquency or default carries through to the augmented specifications. Regarding the newly inserted covariates, in general, the dummies for the various types of equipment do not exhibit a consistent pattern: changes in the way we measure contract performance affect the significance of these dummies. The effect that appears most robust across specifications is the positive correlation between the dummy for copy equipment and the delinquency rate. The size of the contract generally seems to have no role in contract performance. Finally, in Table 6, Panel D, we partition the sample according to the type of contract (lease or loan). The results essentially hold for both categories of contracts. Interestingly, the treatment variable turns out to have a positive effect on the probability of bankruptcy for leases but not for loans.

6.4. Sorting by firm attributes

A possible way to better identify the effect of lenders’ information pooling on borrowers’ performance is to test whether this effect is stronger for firms that are reputed to be more informationally opaque or riskier. Small and young firms are allegedly informationally opaque because they lack an established track record accessible by lenders (Petersen and Rajan, 1994); in turn, firms with low credit ratings are a natural candidate for risky businesses. In light of these considerations, we now reestimate the model in (1) by sorting the firms according to their size, age, and rating. We first restrict attention to the subsample of firms for which we have demographic information and partition this subsample on the basis of firm size and age. The results are displayed in Table 6. In Panel A, we experiment by dropping contracts granted to firms with over 100 or over 500 employees. In line with our null hypothesis, the results reveal that the impact of *information* on serious delinquencies is generally stronger for the subsamples of small and medium-sized firms than for the full subsample for which we have information on the number of employees. For example, for the average contract lender entry into the bureau translates into a drop of the number of times over 90 days past due from 0.424 to 0.332 for businesses with no more than 100 employees versus a drop from 0.406 to 0.336 for all firms (see Panel A, column 3).²¹ Next, in Panel B we sort the firms according to their age. The improvement in contract performance induced by lender entry into the bureau appears to be stronger for young firms (below the first quartile of age) regardless of whether we use measures of moderate delinquency such as the maximum number of days past due or indicators of major events such as the number of times over 90 days past due.

Finally, we sort the firms according to their rating. As is typically the case, PayNet does not disclose the

²⁰ Possible types of equipment are agricultural, auto, bus, construction equipment, computers, copy equipment, forks, medium size trucks, manufacturing, no equipment, retail, telecommunications, trucks, unknown, vending machines, and waste processing. For guarantor, the options are yes, no, corporate, personal, both, and unknown. For payment frequency, the options are monthly, quarterly, semiannually, annually, or other. Notice that information on some characteristics is

(footnote continued)

unavailable for some contracts so that in the regressions in Table 5 there are less observations than in the baseline regressions.

²¹ Clearly, the reader should interpret these results with caution because, unlike for firm ratings and age, we observe the number of employees only at the end of the sample (2007).

Table 5

Information sharing and contract performance—further contract attributes.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on moderate delinquencies and major events of delinquency, default, and bankruptcy after controlling for further contract attributes. Average days past due (DPD) and maximum DPD are, respectively, the average and the maximum number of days payments were late during the contract (both measured in logs). Number of times over 30 DPD and number of times over 90 DPD are variables capturing, respectively, the (log) number of times that payments on the contract were 30 or more days late and the (log) number of times that payments on the contract were 90 or more days late. Serious delinquency is an indicator variable taking the value of one if there was at least one payment 90 or more days late during the contract, zero otherwise. Major event of default is an indicator variable taking the value of one if during the contract a major event of default (e.g., a legal action or a debt collection) occurred, zero otherwise. Bankruptcy is an indicator variable taking the value of one if during the contract the firm went bankrupt, zero otherwise. Information is a dummy variable equal to one if the contract began after lender entry into the PayNet bureau, zero otherwise. Contract maturity is measured in months and contract size is measured in millions of dollars. All other characteristics are captured by dummies which equal one if the contract has the specified property, zero otherwise. All regressions include lender dummies, firm dummies, and time dummies controlling for both the start date and the close date of the contract. Sample details are in Table 2. For each dependent variable, we report ordinary least squares (OLS) estimates. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. For all the regressions, the table also reports the adjusted *R*-squared.

	Average DPD	Maximum DPD	Number of times over 30 DPD	Number of times over 90 DPD	Serious delinquency (yes/no)	Major event of default	Bankruptcy
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)
Information	−0.108*** (0.037)	−0.277*** (0.060)	−0.050* (0.026)	−0.050** (0.021)	−0.021** (0.010)	−0.005** (0.002)	0.009** (0.003)
Contract maturity	0.002* (0.001)	0.002** (0.001)	0.001*** (0.0005)	0.001** (0.0003)	0.001*** (0.0002)	0.0001* (0.0001)	0.0004*** (0.0001)
Loan dummy	−0.123*** (0.046)	−0.124* (0.071)	−0.040 (0.029)	−0.023* (0.014)	−0.009 (0.009)	0.003** (0.010)	−0.0006 (0.003)
Contract size	1.33e−9 (1.43e−8)	−1.72e−8 (2.43e−8)	−5.26e−10 (1.19e−8)	−9.96e−9** (4.96e−9)	−8.9e−9*** (3.41e−9)	−1.27e−10 (4.03e−10)	−7.86e−10 (1.37e−9)
Guarantor	−0.061** (0.026)	−0.056 (0.04)	−0.003 (0.018)	−0.021** (0.010)	−0.020*** (0.006)	−0.005*** (0.002)	0.001 (0.003)
Monthly freq.	0.011 (0.038)	−0.063 (0.052)	−0.063*** (0.022)	−0.005 (0.012)	−0.014 (0.009)	−0.014*** (0.003)	0.003 (0.003)
Quarterly freq.	−0.141 (0.122)	−0.305* (0.176)	−0.392*** (0.072)	0.103** (0.053)	0.098** (0.042)	−0.021* (0.012)	0.017 (0.014)
Annual freq.	−0.182** (0.091)	−0.406** (0.171)	−0.254*** (0.054)	0.025 (0.037)	0.002 (0.019)	−0.018** (0.007)	−0.004 (0.006)
Equipment type	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	14,656	18,096	18,096	18,096	18,096	18,096	18,096
Adj <i>R</i> -sq	0.648	0.617	0.689	0.570	0.583	0.629	0.633

proprietary method used to calculate ratings. However, the reports of PayNet suggest that this method is very close to that used by rating agencies. The method puts special emphasis on historical serious delinquencies, recent delinquencies, and borrowings paid down. Other factors mentioned in the reports include total indebtedness, scheduled payments, age of the accounts, length of the credit track record, and some demographic characteristics of the borrower. In general, the results in Panel C do not hint at a stronger effect of information sharing on the performance of contracts of low rating businesses.

Besides subsampling by firm characteristics, we experimented by interacting *information* with the number of employees, and with firm age and rating (these estimates are not tabulated to conserve space but are available upon request). The estimates yield the same insights as those obtained by partitioning the sample.

6.5. Timing issues

In the baseline specification, our definition of the treatment variable is based on the premise that lenders'

information sharing affects contract performance if and only if information is available before the start of the contract. However, one could conjecture that the arrival of additional information could improve borrowers' repayment performance even in contracts that are already open when the lender joins the bureau. This can occur because the news that the lender has joined the bureau could discipline the borrower (see point iii. in Section 4). In fact, the borrower would now be aware that any delinquency or default during the remaining part of the contract will be recorded by the bureau, influencing his future ability to obtain credit (Padilla and Pagano, 2000). Clearly, whether this scenario is plausible depends on whether relevant choices are made during the life of a contract that can still significantly affect its performance or whether, instead, the choices that really matter are made before the start of a contract; it also depends on whether firms with contracts already open at the time of lenders' entry into the bureau became quickly aware of this entry.

To account for these issues, we reestimate the model in (1) with an alternative definition of the *information* dummy. In particular, we experiment with defining the

Table 6

Information sharing and contract performance—subsampling by firm and contract attributes.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on moderate delinquencies and major events of delinquency, default, and bankruptcy obtained by sorting firms according to their number of employees (Panel A), age (Panel B), rating (Panel C), and by sorting contracts according to the type of deal (loan or lease, Panel D). The number of employees is reported as of 2007, while age and rating refer to the time when the contract began. Since information on employees, age, and rating is missing for a portion of our sample, we rerun the basic regressions including only contracts of firms for which these characteristics are known, and report these results under the label 'All.' In Panel A, the label ' ≤ 100 ' refers to estimates from regressions including only firms with 100 employees or less, and similarly for the label ' ≤ 500 .' In Panel B, the label ' ≤ 5.58 ' refers to regressions including contracts of firms in the bottom quartile of age (as of the contract start date), and ' ≤ 11.68 ' to contracts of firms below the median age as of the contract start date. In Panel C, 30 and 55 represent the ratings at the 25th percentile and at the median. Average days past due (DPD) and maximum days past due (DPD) are, respectively, the average and the maximum number of days payments were past due during the contract (both measured in logs). Number of times over 90 DPD is a variable capturing the (log) number of times that payments on the contract were 90 or more days late. Major event of default is an indicator variable taking the value of one if during the contract a major event of default (e.g., a legal action or a debt collection) occurred, zero otherwise. Bankruptcy is an indicator variable taking the value of one if during the contract the firm went bankrupt, zero otherwise. Information is a dummy variable that takes the value of one if the contract began after lender entry into the PayNet bureau, zero otherwise. All regressions include lender dummies, firm dummies, and time dummies controlling for both the start date and the close date of the contract. All regressions also control for contract maturity (in months) and the regressions in Panels A–C also control for contract type (a Loan dummy variable which takes the value of one if the contract is a loan, zero otherwise). Sample details are in Table 1. For each dependent variable, we report ordinary least squares (OLS) estimates. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

Panel A: Subsampling by number of employees							Panel B: Subsampling by firm age						
		Average DPD	Maximum DPD	Number of times over 90 DPD	Major event of de- fault	Bankruptcy			Average DPD	Maximum DPD	Number of times over 90 DPD	Major event of default	Bankruptcy
		(1)	(2)	(3)	(4)	(5)			(1)	(2)	(3)	(4)	(5)
All	Information	−0.089** (0.042)	−0.190*** (0.066)	−0.051** (0.020)	−0.008*** (0.003)	0.002 (0.004)	All	Information	−0.124*** (0.042)	−0.247*** (0.069)	−0.086*** (0.023)	−0.009*** (0.003)	0.002 (0.004)
	Contract mat.	0.004*** (0.001)	0.006*** (0.001)	0.002*** (0.0004)	0.0002** (0.0001)	0.0003*** (0.0001)		Contract mat.	0.004*** (0.001)	0.006*** (0.001)	0.001*** (0.0004)	0.00002 (0.0001)	0.0002** (0.0001)
≤ 100	Information	−0.061 (0.047)	−0.176** (0.075)	−0.067*** (0.024)	−0.010*** (0.003)	−0.002 (0.005)	≤ 5.58	Information	−0.057 (0.109)	−0.431*** (0.163)	−0.131** (0.062)	−0.003 (0.002)	0.006 (0.012)
	Contract mat.	0.005*** (0.001)	0.008*** (0.001)	0.002*** (0.0004)	0.0002** (0.0001)	0.0002* (0.0001)		Contract mat.	0.004* (0.003)	0.001 (0.003)	0.002 (0.001)	−0.0002 (0.0001)	−0.0001 (0.0002)
≤ 500	Information	−0.089** (0.043)	−0.172** (0.068)	−0.054*** (0.020)	−0.009*** (0.003)	0.002 (0.004)	≤ 11.68	Information	−0.031 (0.065)	−0.278** (0.105)	−0.078** (0.038)	−0.005* (0.003)	0.002 (0.007)
	Contract mat.	0.004*** (0.001)	0.006*** (0.001)	0.002*** (0.0004)	0.0002** (0.0007)	0.0003*** (0.0001)		Contract mat.	0.005*** (0.001)	0.006*** (0.002)	0.001* (0.0007)	−8.00e−06 (0.0001)	−8.42e−7 (0.0001)
Panel C: Subsampling by firm rating							Panel D: Subsampling by contract type						
		Average DPD	Maximum DPD	Number of times over 90 DPD	Major event of default	Bankruptcy			Average DPD	Maximum DPD	Number of times over 90 DPD	Major event of default	Bankruptcy
		(1)	(2)	(3)	(4)	(5)			(1)	(2)	(3)	(4)	(5)
All	Information	−0.027 (0.043)	−0.258*** (0.073)	−0.069** (0.028)	−0.008** (0.003)	0.011** (0.005)	All	Information	−0.114*** (0.033)	−0.232*** (0.054)	−0.047*** (0.017)	−0.004** (0.002)	0.008** (0.003)
	Contract mat.	0.002** (0.001)	0.002 (0.001)	0.001*** (0.0004)	0.0001 (0.0001)	0.0005*** (0.0001)		Contract mat.	0.003*** (0.001)	0.004*** (0.001)	0.001*** (0.0003)	0.0001* (0.00005)	0.0004*** (0.0001)

≤ 30	Information	0.136 (0.157)	–0.081 (0.199)	–0.027 (0.071)	–0.023 (0.015)	0.021 (0.014)	Loan	Information	–0.079* (0.043)	–0.259*** (0.094)	–0.157*** (0.043)	–0.005 (0.004)	0.002 (0.003)
	Contract mat.	0.002 (0.003)	–0.002 (0.003)	0.004** (0.001)	0.001* (0.0004)	0.001*** (0.0003)	Contract mat.	Contract mat.	0.001 (0.001)	0.001 (0.002)	0.0003 (0.0003)	0.0002** (0.0001)	0.0004*** (0.0001)
≤ 55	Information	0.010 (0.093)	–0.256*** (0.128)	–0.048 (0.054)	0.016* (0.008)	0.004 (0.010)	Lease	Information	–0.206*** (0.053)	–0.278*** (0.074)	0.001 (0.018)	–0.005** (0.002)	0.013*** (0.006)
	Contract mat.	0.003 (0.002)	0.001 (0.002)	0.003*** (0.001)	0.0003 (0.0002)	0.0007*** (0.0002)	Contract mat.	Contract mat.	0.005*** (0.001)	0.008*** (0.001)	0.002*** (0.0004)	0.00002 (0.0001)	0.0005*** (0.0001)

treatment variable as a dummy that equals one if the contract ended after lender entry into the bureau and started no earlier than one month, three months, six months, or nine months before the date of entry. The results, reported in Table 7, Panel A, essentially carry through after this change in the construction of the treatment variable, but, expectedly, the estimated coefficient tends to drop in magnitude and progressively lose statistical significance as we move back the threshold date. For instance, if we consider the maximum number of days past due as the dependent variable, we estimate a coefficient on *information* of –0.192 when we shift the threshold date back by one month, –0.166 when we shift it back by six months, and a non-significant coefficient when we shift it back by nine months.

An argument that goes in a direction opposite of that just discussed is that not only entry into the bureau could have no impact on contracts already open but also that it could affect borrowers' repayment performance with a small lag. For example, although lenders had time to prepare before their entry into the bureau, the reader may argue that after its entry, a lender needed to refine the protocol for processing the new information. In Table 7, Panel B, we then reestimate the regressions by redefining *information* as a dummy that takes the value of one if the contract started no earlier than one month, two months, three months, or six months after the lender's entry date, zero otherwise. Typically, we detect the strongest and most significant effects of (our proxy for) information sharing when we employ either the baseline specification or the specification in which the treatment variable equals one for contracts started at least one month after the lender's entry, with the effect progressively weakening as the threshold date is pushed further ahead. This result suggests that the affiliation to the bureau improves borrowers' repayment performance either as soon as information becomes available or with a short lag.

6.6. Other robustness issues

A point that deserves closer scrutiny regards the actual use of the PayNet database by lenders. PayNet members pay a very low fee to retrieve credit reports about applicants. This could raise some issues. First, it could be that for some contracts a lender chose not to use the database when making its lending decision. However, if there is a bias, this would be towards finding no effect of the affiliation to the bureau on contract performance so that we do not risk overestimating the impact of lenders' information sharing. Second, the reader could wonder whether the decision to retrieve a credit report could be influenced by the benefit expected from using the report. However, the fees requested to retrieve reports are very low. Moreover, these fees are typically prepaid as lenders buy the right to retrieve a given number of PayNet reports at the beginning of the year. These features make it very unlikely that the use of a credit report is endogenous to the perceived quality of the borrower.

Another possible concern is that the data collected before lenders joined the bureau might be of different quality from the data collected after lenders joined it.

Table 7

Information sharing and contract performance—timing issues.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on contract performance allowing affiliation to become effective either with a lead (i.e., on contracts already open—Panel A) or with a lag (i.e., on contracts starting at least some time after the lender entry into the bureau—Panel B). For example, column $-6m (+3m)$ displays estimated coefficients for a dummy variable that is equal to one if the contract start date is no earlier than the lender's date of entry into the bureau minus six months (plus three months) and the close date is after lender entry, zero otherwise. For all the dependent variables, we estimate ordinary least squares (OLS) regressions with firm dummies, lender dummies and time dummies, controlling for both the start date and the close date of a contract. Average days past due (DPD) and maximum days past due (DPD) are, respectively, the average and the maximum number of days payments were past due during the contract (in logs). Number of times over 90 DPD is a variable capturing the (log) number of times that payments on the contract were 90 or more days late. Serious delinquency is an indicator variable taking the value of one if there was at least one payment 90 or more days late during the contract, zero otherwise. Major event of default is an indicator variable taking the value of one if during the contract a major event of default (e.g., a legal action or a debt collection) occurred, zero otherwise. Bankruptcy is an indicator variable taking the value of one if during the contract the firm went bankrupt, zero otherwise. Sample details are in Table 1. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Impact on open contracts					
	−9m	−6m	−3m	−1m	0
	(1)	(2)	(3)	(4)	(5)
Average DPD	−0.102*** (0.035)	−0.125*** (0.033)	−0.087*** (0.032)	−0.107*** (0.033)	−0.114*** (0.033)
Maximum DPD	−0.013 (0.057)	−0.166*** (0.050)	−0.135*** (0.050)	−0.192*** (0.052)	−0.232*** (0.054)
No. of times over 90 DPD	−0.009 (0.013)	−0.004 (0.013)	0.006 (0.013)	−0.033** (0.017)	−0.047*** (0.017)
Serious delinquency (yes/no)	−0.008 (0.008)	−0.007 (0.009)	−0.001 (0.008)	−0.018** (0.009)	−0.024*** (0.009)
Major event of default	0.007*** (0.003)	0.001 (0.003)	0.001 (0.003)	−0.004* (0.002)	−0.005** (0.002)
Bankruptcy	−0.008** (0.004)	−0.010** (0.004)	−0.0003 (0.003)	0.006* (0.003)	0.008** (0.003)
Panel B: Lagged impact on contracts					
	0	+1m	+2m	+3m	+6m
	(1)	(2)	(3)	(4)	(5)
Average DPD	−0.114*** (0.033)	−0.133*** (0.034)	−0.103*** (0.034)	−0.068** (0.033)	−0.041 (0.030)
Maximum DPD	−0.232*** (0.054)	−0.237*** (0.055)	−0.221*** (0.055)	−0.166*** (0.054)	−0.067 (0.052)
No. of times over 90 DPD	−0.047*** (0.017)	−0.031* (0.017)	−0.017 (0.017)	−0.004 (0.016)	0.038*** (0.014)
Serious delinquency (yes/no)	−0.024*** (0.009)	−0.014 (0.009)	−0.008 (0.009)	−0.004 (0.009)	0.014* (0.008)
Major event of default	−0.005** (0.002)	−0.004** (0.002)	−0.003 (0.002)	−0.001 (0.002)	0.003* (0.002)
Bankruptcy	0.008** (0.003)	0.009*** (0.003)	0.009*** (0.003)	0.012*** (0.003)	0.005* (0.003)

In particular, one could wonder whether the data that predate the inception of the bureau are somewhat less accurate. However, we do not believe this to be the case. The documents of ELFA stress the high quality of the PayNet database and, in particular, that the data on customers' payment histories (including those on old contracts) were directly supplied from lenders' computerized systems. There is no reason to believe that in these computerized systems there were more mistakes before the inception of the bureau. In fact, as reported explicitly by PayNet, the bureau required an interface with the existing computer systems of the lenders but it did not interfere with the recording of contracts.

Finally, one could wonder whether changes in payment methods occurred during this period and whether

the observed reductions in delinquencies and defaults also reflect the introduction of more efficient payment methods. However, such a structural shock in the payment technology would affect all lenders more or less at the same time and would then be picked up by the time dummies. Moreover, to further assuage concerns, we checked whether new payment methods were introduced in the years of the sample and did not find evidence of such a structural change in the equipment finance industry.

6.7. Further tests

In this section, we perform a battery of further tests (the results are gathered in supplementary tables

available from the authors). These tests have two objectives. The first is to better disentangle the mechanisms through which lenders' information sharing improved the performance of the portfolios of bureau members. The second is to examine the effects of the bureau on the credit market at large, including equipment finance companies outside the bureau.

6.7.1. Effects on bureau members. Intensive and extensive margins

Lenders' information sharing can improve borrowers' repayment performance by mitigating moral hazard or adverse selection problems. Lower moral hazard should result in an improvement in firms' repayment performance for a given pool of borrowers (intensive margin effect); lower adverse selection should result in an improvement in the quality of the pool of borrowers (extensive margin effect). To try to disentangle the intensive margin effect, we reestimated the model in (1) by restricting attention to firms that were financed by a lender both before and after the lender's entry into the bureau. As in the baseline specification, the treatment variable takes the value of one if the start date of the contract is after the lender's entry into the bureau, zero otherwise. The results are in line with those obtained for the full sample, suggesting that the intensive margin contributes to the positive impact of information sharing on borrowers' performance. For example, for the average number of days past due, the estimated coefficient on the treatment variable implies that for the average contract lender entry into the bureau triggers a drop in this number from 5.59 to 5.05. For the average contract, lender entry into the bureau also implies a reduction from 7.3% to 5.7% in the probability of a serious delinquency.

Next, we turn to study the extensive margin effect. For this purpose, we reestimated the model in (1) by restricting attention to the first contract that appears to be stipulated by a firm in the sample (if more than one contract was stipulated on the same date, we considered all of them). We estimated two sets of regressions. In one, we considered firms that appear to have been funded only before or only after the entry of a lender into the bureau; in the other, we considered all firms. Overall, the estimates suggest that the extensive margin plays an even stronger role than the intensive margin in the impact of information sharing on moderate delinquencies. For example, considering the first set of regressions, the estimated coefficient on *information* implies that lender entry into the bureau makes the average number of days past due drop from 9.74 to 7.08 for the average contract. Interestingly, these findings could match the result of Berger, Frame, and Ioannidou (2011) that the availability of collateral tends to mitigate moral hazard more than adverse selection and, hence, that in an industry with wide availability of collateral like the equipment finance industry, lenders' information sharing retains a stronger role in mitigating adverse selection. However, we do not find evidence that the treatment variable affects (measures of) major events of delinquency or default along the extensive margin: the estimated coefficient on *information* is not significant at conventional levels.

6.7.2. Effects on the credit market at large

The reader may wonder what effects the bureau had on the credit market at large. First of all, one may wonder whether some firms "substituted" delinquencies and defaults on equipment finance contracts with delinquencies and defaults on other categories of financing contracts (which we do not observe in our data). However, even if some borrowers behaved this way, it would not diminish the relevance of the results. In fact, it would indicate that sharing information in equipment finance allows lenders to alleviate moral hazard or perform effective screening of firms in this financing activity, so that indeed, information sharing has a beneficial impact. This observation suggests that, to minimize the risk that borrowers associate their better credit performance on equipment finance contracts with a worse performance on other segments of credit contracts, financiers should make their information sharing more comprehensive. A second consideration is that equipment investment and financing constitute a large portion of the total expenditures and debt of firms, respectively. Thus, a borrower's margin for substituting delinquencies and defaults on equipment finance contracts with delinquencies and defaults on other contractual obligations might be limited. This is especially true if we account for the size of the firms in our sample together with the lumpy nature of most equipment investment. For example, the purchase of a truck is likely to absorb a sizeable portion of the operating expenses of a small transportation company with, say, one or two dozen employees.

A different but related issue is whether, inside the equipment finance industry, firms seeking financing switched from lenders that joined the bureau to lenders that remained outside the bureau. There are two questions relevant for this point. First, whether indeed some equipment finance companies chose not to join the PayNet bureau and, if so, why. Second, what implications this could have had for the impact of the bureau. Starting from the latter question, the theoretical literature predicts that the presence of lenders outside a bureau might dilute the disciplining effect of information sharing. Regarding borrowers' moral hazard, borrowers' incentive to misbehave can be higher if they know that, despite their misbehavior, there will be lenders outside the bureau unaware of this and ready to provide funding (Padilla and Pagano, 2000).²² Regarding adverse selection, a prediction of some theoretical studies is that in a credit market affected by adverse selection, a pool of lenders that share information can "cherry pick" the best clients and leave uninformed lenders with the worst borrowers, thus also hindering their participation to the credit market (Dell'Ariccia, Friedman, and Marquez, 1999). Hence, lenders outside the bureau might experience a reduction in the quality and performance of their portfolios and in their lending activity. The data set allows us to test for the presence of such an effect. We split the lenders in the

²² The theoretical models do not imply that the presence of uninformed lenders outside the bureau should affect lenders' hold-up. In fact, the presence of multiple lenders inside the bureau should suffice to generate competition and prevent hold-up problems.

sample into “early joiners” (lenders that entered the bureau before the median date of entry) and “late joiners” (lenders that entered the bureau after the median date of entry). We then carried out an exogeneity test similar to that illustrated in Section 6.1 separately for the groups of early joiners and late joiners, with pre-entry and post-entry time dummies. The estimated coefficients on the pre-entry and post-entry time dummies suggest no significant pattern in the portfolio performance of late joiners in the pre-entry period and of early joiners in the post-entry period. Thus, there appears to be no evidence of spillover effects between early and late joiners.

Let us now turn to the first question, that is, how many equipment finance companies remained outside the bureau. While we lack exact information on this, PayNet and ELFA stress that the bureau was very popular in the industry. In particular, a report of ELFA (ELT, 2002) documents that within four years from the inception of the bureau, eight of the ten largest lenders in the equipment finance industry reached an agreement with PayNet for joining the bureau. More in general, the report emphasizes that, thanks both to the official endorsement of ELFA and to the characteristics of PayNet, the bureau experienced almost universal adoption in the industry. That said, it is useful to consider what could have induced a lender to remain outside the bureau. Competition is probably the main factor. If in the credit market competition is intense, a lender might be concerned that sharing its proprietary information with competitor lenders will cause a loss of market shares. Studying an experimental credit market, Brown and Zehnder (2010) find that the effect of competition is significantly smaller than the benefit of information sharing due to lower adverse selection. These findings suggest that equipment finance companies might not be too worried about losing clients to other lenders if they share their information. Furthermore, according to the reports of ELFA (see, e.g., ELT, 2002), the fact that the PayNet bureau is a “closed” network mitigates the effect of competition and the associated incentive to remain outside the bureau. In fact, a lender knows that it will not obtain information if, in turn, it does not release its own proprietary information. This gives a lender a strong incentive to join the bureau.

7. Evidence from contract dynamics

In the tests of the previous section, we treated each contract as a single observation. This allowed us to exploit the size of the data set in full. A drawback of this approach is that, by collapsing all the information on the performance of a contract into a unique observation, we may lose precision in controlling for the impact on contract performance of time-varying factors, such as macroeconomic conditions. This could be especially problematic for very long-term contracts, during which macroeconomic conditions can change more substantially. Moreover, one can probably gain interesting insights from investigating the impact of information sharing on the internal dynamics of contracts. In this section, we perform tests in which we use information at quarterly frequency on

delinquencies occurring during the life of contracts. We also employ a duration analysis to investigate the time to first delinquency. These tests entail a loss of observations because before the first quarter of 2000, we do not observe the (lack of) repayment on a quarter-by-quarter basis. Thus, we can carry out the tests of this section only on a subset of our sample.

7.1. The empirical models and results

Our methodological approach broadly parallels that followed in the previous section. We construct measures of delinquency for each contract on a quarterly basis and test whether the affiliation of the lender to the bureau affects these measures. The data set provides information on 146,430 contract-quarter observations. In Panel A of Table 8, columns 1–4, we take a first glance at the impact of lenders’ information exchange by estimating a model similar to (1). We use contract-quarter number of days past due ($DPDQ$) as the dependent variable and accordingly redefine the time dummies. Moreover, we account for the fact that over the course of a contract the delinquency of a borrower in a quarter could be correlated with her delinquency in the previous quarter(s). This suggests that lenders’ information sharing could also influence the persistence of delinquencies during the contract. We then estimate the following model:

$$\begin{aligned} DPDQ_{ijst} = & c + \rho DPDQ_{ijst-1} + \alpha Information_{js} \\ & + \beta (Information_{js} DPDQ_{ijst-1}) + \sum_j \gamma_j D_j^L \\ & + \sum_i \delta_i D_i^F + \sum_t \varepsilon_t D_t^P + \zeta C_{ijs} + \eta M_{ijs} + u_{ijst}, \end{aligned} \quad (2)$$

where $DPDQ_{ijst}$ is the number of days past due in quarter t in a contract between firm i and lender j that started on date s , $DPDQ_{ijst-1}$ is the lagged dependent variable, D_j^L is a lender fixed effect, D_i^F is a firm fixed effect, D_t^P is a quarter time dummy, C_{ijs} is a dummy that takes the value of one if the contract is a loan or a related contract, M_{ijs} is the maturity of the contract, and, in a first specification, $information_{js}$ is an indicator variable taking the value of one if the lender was affiliated to the bureau when the contract started, zero otherwise. In model (2), the β coefficient on the interaction term between the lagged dependent variable and the treatment variable reflects the impact of lenders’ information sharing on the persistence of delinquencies during the contract.

Column 2 of Panel A reports estimation results. Both the estimated coefficient α on *information* and the estimated coefficient β on the interaction term are significantly negative. Thus, the affiliation to the credit bureau appears not only to reduce arrears throughout the life of a contract but also to reduce the persistence of delinquencies over the life of the contract. In column 1 of the same panel, we reestimate the model allowing the affiliation of a lender to the bureau to have an impact even if the contract was already open at the time of the treatment. Precisely, we now define *information* as a dummy that takes the value of one if the contract-quarter observation

Table 8

Information sharing and contract dynamics.

This table reports coefficient or hazard ratio estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on contract delinquencies using contract-quarter observations. In the regressions of Panel A, the dependent variable is the number of days past due (measured in logarithms) in a given quarter for a given contract. Panel B presents the results of a duration analysis for the time to occurrence of the first delinquency in a contract. In columns 2, 4, and 6 of Panel A and in columns 1 and 2 of Panel B, information is a dummy variable that takes the value of one if the contract began after lender entry into the PayNet bureau, zero otherwise. In columns 1, 3, and 5 of Panel A, information is a dummy variable that takes the value of one if the lender was in the PayNet bureau in the quarter observed, zero otherwise. The regressions in Panel A include the lagged dependent variable as well as its interaction with the information dummy. Sample details are in Table 1. Regressions 1 and 2 are estimated by ordinary least squares (OLS), regressions 3 and 4 are estimated by Tobit, regressions 5 and 6 are estimated with the Arellano and Bond (1991) generalized method of moments (GMM) approach. Regressions 1 and 2 of Panel B are estimated with the Cox model (regression 1 without firm dummies, regression 2 with firm dummies). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. In Panel A, for the Arellano and Bond regressions, the table also reports the results of a Wald test while for all the other regressions, the table reports the adjusted *R*-squared (pseudo if Tobit). In Panel B, the table reports the results of an inference test that all model parameters are zero.

Panel A: Contract-quarter observations						
	OLS (1)	OLS (2)	Tobit (3)	Tobit (4)	Arellano and Bond	
					(5)	(6)
Information	−0.015 (0.016)	−0.031*** (0.009)	−0.514*** (0.183)	−0.379*** (0.118)	0.361*** (0.028)	–
Lagged dependent	0.398*** (0.008)	0.371*** (0.005)	1.283*** (0.042)	1.372*** (0.030)	0.236*** (0.006)	0.262*** (0.006)
Interaction	−0.042*** (0.008)	−0.016** (0.008)	0.257*** (0.044)	0.275*** (0.039)	−0.631*** (0.007)	−0.668*** (0.008)
Contract maturity	0.00004 (0.0003)	−0.0001 (0.0003)	0.004 (0.003)	0.004 (0.003)		
Loan dummy	−0.051*** (0.015)	−0.052*** (0.016)	−0.187 (0.127)	−0.190 (0.127)		
Firm dummies	Yes	Yes	No	No	Yes	Yes
Effect on open contracts	Yes	No	Yes	No	Yes	No
No. of observations	146,430	146,430	146,430	146,430	123,113	123,113
Adj. <i>R</i> -sq (pseudo <i>R</i> -sq if Tobit)	0.476	0.476	0.150	0.150		
Wald test					9,990.49	8,972.68

Panel B: Duration analysis		
	Cox model (1)	Cox model (2)
Information (hazard ratio)	0.706*** (0.067)	0.768*** (0.040)
Contract maturity	1.003** (0.001)	0.998** (0.001)
Loan dummy	0.714*** (0.055)	0.731*** (0.036)
Firm dummies	No	Yes
No. of observations	27,765	27,765
Inference test	(Wald) 1,777.38***	(Likelihood ratio) 11,784.56***

is after the lender's entry into the bureau, zero otherwise. The insights we draw from the estimates are essentially unaltered, although the statistical significance of the coefficient on the treatment variable tends to drop.

There are two fundamental problems associated with the empirical model in (2). The model hinges on the assumption that any heterogeneity across contracts is fully captured by lenders' affiliation to the bureau and by the contract attributes added to the specification. Clearly, the reader may wonder whether we are omitting contract-level factors that are relevant in determining delinquencies. As shown by Wooldridge (2008), for example, putting an individual contract effect into the error term is a source of inconsistency of the ordinary least square estimates if this effect is correlated with other regressors. To tackle this issue, we can eliminate the unobservable, individual contract effect by taking the first

difference of (2), that is

$$\Delta DPDQ_{ijst} = \rho \Delta DPDQ_{ijst-1} + \gamma \Delta Information_{js} + \zeta \Delta (Information_{js} DPDQ_{ijst-1}) + \sum_t \delta_t D_t^p + \Delta u_{ijst}. \quad (3)$$

Given the presence of a lagged dependent variable in the original model, however, estimating the transformation in (3) by ordinary least squares would still produce inconsistent estimates. This would occur because the first-differenced error term Δu_{ijst} is correlated with the first-differenced lagged dependent variable $\Delta DPDQ_{ijst-1}$. Although the size of the inconsistency can be small for a sufficiently large time span of the series, we cannot neglect this issue. Therefore, we resort to an instrumental variable approach. Arellano and Bond (1991) propose a generalized method of moments (GMM) to estimate a first-difference model such as (3). They

demonstrate that using lags of both the dependent and independent variables as instruments delivers consistent and asymptotically efficient parameter estimates, even if the panel is unbalanced as in our case. Before turning to the regression output, a caveat is in order regarding the first-differenced $\Delta information_{js}$ term in (3). This term is meaningful only when we allow the entry of a lender into the bureau to have an impact on contracts already open. By contrast, in the specifications in which we only allow entry to exert a role in contracts not yet started, we are unable to disentangle the effect of $\Delta information_{js}$ (which would have the constant value of zero). In these specifications, we can only estimate the impact of lenders' information sharing on the persistence of delinquencies during contracts, as captured by the β coefficient on the interaction term.

The estimation results from the Arellano and Bond model are shown in columns 5 and 6 of Panel A in Table 8. Consider the case in which we do not allow a lender's affiliation to the bureau to have an effect on contracts already open at the time of entry into the bureau (column 6). Under this specification, the estimated coefficient on the lagged dependent variable takes a positive value (p -value < 0.01). Most importantly, affiliation to the bureau appears to reduce the persistence of delinquencies: the interaction term has a negative estimated coefficient, also significant at the 1% level. Let us now turn to the specification in which we allow *information* to affect open contracts. The estimates, displayed in column 5, show again that the coefficient on the $\Delta(information_{js}DPDQ_{ijst-1})$ interaction term is negative and statistically significant while the coefficient on $\Delta information_{js}$ has a positive sign. Summing the two terms at the sample mean of $DPDQ_{ijst-1}$, the overall effect of entry into the bureau on delinquencies is negative, although not large.

We put forth three possible interpretations for the weaker result we obtain when we allow for an effect on contracts already open. A first intriguing interpretation stems from the theoretical literature. The main beneficial effect of information sharing on ongoing contracts should operate through tighter discipline on borrowers: once a borrower becomes aware that his lender is a member of a bureau, he will know that any additional delinquency on his contractual obligations will be recorded by the bureau and alter his future chances to obtain credit. Padilla and Pagano (2000) predict that this effect tends to fade if lenders also pool "white" information on borrowers' characteristics, such as their ratings. In this case, a high quality entrepreneur knows that his quality will be disclosed to lenders anyway, regardless of whether his credit history is good or bad. Since the PayNet bureau reports the rating of borrowers (besides their credit histories), the finding that the effect of information sharing gets diluted if we also allow for an impact on ongoing contracts could then be in line with this argument. A second possibility is that firms concentrate their information acquisition about lenders and credit conditions at the time of contract negotiation, so that the firms that have already signed a contract with a lender could be less aware of the lender's entry into the bureau than the firms that negotiate an agreement after the lender's entry. Finally, the last interpretation relates to the use of

covenants. Information can be helpful to discipline a borrower and prevent his misbehavior especially if it allows the lender to enforce covenants introduced at the start of the contract (Rajan and Winton, 1995). A lender that is not affiliated to a credit bureau when a contract starts could then have little incentive to introduce covenants in the contract. The lack of covenants in contracts already open at the time of entry could help explain why the effect of information sharing on delinquencies weakens when we allow for an impact on such contracts.

7.2. Duration analysis

The availability of detailed information on contract histories allows us to investigate further the performance of borrowers with a duration analysis. In particular, we are interested in testing whether lenders' information sharing affects the time to the occurrence of the first delinquency (a positive number of days past due) during a contract. We evaluate the risk of delinquency using a proportional hazard model and treat contracts in which no delinquency occurs as censored observations.²³ In this setting, the hazard rate $h(t)$ is defined as the probability that the lender faces the first delinquency on the contract at time t , conditional on never having faced a delinquency on the contract prior to that time. In the proportional hazard framework, the instantaneous hazard function takes the form

$$h(t; X) = e^{X\beta} h_0(t), \quad (4)$$

where $h_0(t)$ is the "baseline hazard" (that obtains when all covariates are set to zero), X is a vector of covariates that multiplicatively shift the baseline hazard (without affecting the underlying shape of the hazard function), and β is the vector of parameters to be estimated. In our context, holding everything else constant, the ratio between the predicted hazard for a contract originated after the lender's entry into the bureau and the predicted hazard for a contract originated before entry measures the impact of bureau affiliation on the risk of contract delinquency. Given the proportional hazards assumption, this hazard ratio is independent of time. To analyze the effect of the covariates on the hazard function, we use the Cox model. This is a semiparametric method in that no assumption about the form of the hazard function is made (Greene, 2008). Since the Cox model assumes that the hazard function is continuous while our data are quarterly (and some delinquencies occur at the same survival time or quarter), we adopt the Breslow method (Breslow, 1974) for handling tied delinquencies in the calculation of the log partial likelihood and residuals.

The estimates are in Panel B of Table 8. Contracts originated after lenders joined the bureau have significantly lower hazard rates than contracts originated before entry: the estimated shift in the hazard rate amounts to about 23% (column 2). The estimates also reveal that some lenders have systematically higher hazard rates than others (hazard ratios for lender dummies are not

²³ We do not distinguish between the event of a borrower paying off the loan or lease according to the amortization schedule and the event of the borrower paying it off with an early prepayment.

Table 9

Information sharing and contract terms.

This table reports coefficient estimates and associated standard errors (in parentheses) for the impact of bureau affiliation on contract characteristics (whether there is a guarantor for the contract or not and the contract size). Guarantor is a dummy variable that takes the value of one if there is a guarantor for the contract, zero otherwise. We instead measure contract size as the logarithm of the total dollar amount the firm must pay over the life of the contract. Information is a dummy variable that takes the value of one if the contract began after lender entry into the bureau, zero otherwise. The regressions control for contract maturity (in months) and contract type (a Loan dummy variable which takes the value of one if the contract is a loan, zero otherwise). The regressions also include lender dummies, firm dummies, and time dummies controlling for both the start date and the close date of the contract. Sample details are in Table 1. In regression (1) of both panels we use all firms, in regression (2) we restrict attention to firms below the median rating, in regression (3) to firms below the median age, in regression (4) to firms with less than 100 employees. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. For all the regressions, the table also reports the adjusted *R*-squared.

Panel A: Guarantor				
	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Information	All firms 0.012 (0.012)	Rating ≤ 55 0.068*** (0.021)	Age ≤ 11 0.021 (0.022)	Employees ≤ 100 0.005 (0.018)
Contract maturity	0.001*** (0.0002)	0.001*** (0.0004)	0.001** (0.0004)	0.001*** (0.0003)
Loan dummy	0.037*** (0.011)	0.083*** (0.022)	0.140*** (0.023)	0.006 (0.015)
No. of observations	24,546	8,055	7,701	12,338
Adj <i>R</i> -sq	0.604	0.714	0.597	0.563

Panel B: Contract size				
	OLS (1)	OLS (2)	OLS (3)	OLS (4)
Information	All firms −0.030 (0.029)	Rating ≤ 55 −0.132** (0.056)	Age ≤ 11 0.048 (0.051)	Employees ≤ 100 −0.007 (0.039)
Contract maturity	0.25*** (0.0005)	0.029*** (0.0001)	0.025*** (0.001)	0.026*** (0.006)
Loan dummy	0.149*** (0.025)	−0.011 (0.052)	0.176*** (0.048)	0.170*** (0.032)
No. of observations	27,788	8,946	8,650	13,895
Adj <i>R</i> -sq	0.679	0.740	0.685	0.672

tabulated). Overall, the duration analysis confirms the findings in the rest of the paper that lenders' information sharing improves borrowers' performance.

8. Lending practices

In addition to details on repayment histories, our data set provides information on whether the contracts are backed by a guarantor (personal, corporate, or both). Thus, besides testing its impact on borrowers' repayment performance, we can take a first step towards evaluating whether information pooling changes lending practices. In the sample, about 35% of the contracts are guaranteed. We construct a dummy variable that takes the value of one if the contract is guaranteed, zero otherwise, and replace the dependent variable in model (1) with this *guarantor* dummy. The estimates obtained using a linear probability model suggest that the treatment variable has no significant effect on the probability that the contract has a guarantor (see Table 9, Panel A, column 1). Moreover, if we restrict attention to risky (low rating) businesses, the estimated coefficient becomes positive and statistically significant (column 2). A possible interpretation for this finding is that third parties are more willing to

guarantee a contract when they know that the lender can monitor and discipline the borrower, as is the case if the lender has access to the information contained in a bureau. In fact, in such a case the guarantors expect that the borrower will fulfill his contractual obligations with a higher probability and, hence, they will be less likely to become liable.

A second contractual characteristic on which we have some information is the size of the deal. The regressions in Panel B of Table 9 suggest that lenders' entry into the bureau has no effect on the (natural logarithm) of the deal size (column 1). If instead we restrict attention to low rating firms, we estimate a statistically significant negative effect (column 2). However, since in our data the deal size reflects the amount a firm has to repay, not the amount borrowed, we cannot distinguish interest from principal and separate the impact of information sharing on the cost of borrowing from its impact on the amount borrowed.²⁴

²⁴ The amount borrowed is tied to the cost of the equipment purchased, which is in turn affected by the technological needs of the firm. The reader could then conjecture that it is unlikely that firms buy cheaper machinery after lender entry into the bureau. Under this

9. Conclusion

In this paper, we investigate the consequences of lenders' information sharing using unique contract-level data from a credit bureau that serves the U.S. equipment finance industry. The staggered entry of lenders into the bureau, in conjunction with the small and medium size of borrowing firms, offer a suitable natural experiment to identify the effect of lenders' improved access to information. Consistent with extant theories (especially Pagano and Jappelli, 1993, and Padilla and Pagano, 1997, 2000), we uncover evidence that lenders' information exchange has a beneficial effect on the repayment behavior of firms, reducing the incidence of delinquencies and defaults. This effect appears to be stronger for firms that are reputed to be less informationally transparent, such as small and young firms. The results also suggest that the effects of information sharing are not necessarily homogeneous across all measures of repayment performance. For example, they hint at the possibility that by contrasting delinquencies, information sharing reduces borrowers' flexibility in making regular payments, possibly forcing some (albeit few) of them to resort to bankruptcy to gain time. Finally, we find that lenders' entry into the credit bureau does not reduce the use of guarantees, suggesting that information sharing does not lead financiers to loosen lending standards.

The analysis leaves interesting issues open for future research. The widespread availability of collateral in the equipment finance industry suggests caution in applying our results to industries in which collateral is less diffused. At the same time, this could also imply that our results underestimate the effect of information sharing on contract performance, because the effect could be diluted by the availability of collateral. Clearly, more work is needed to understand the interaction between information sharing and (the use of) collateral. Finally, an aspect that we have not been able to address is the impact of information exchange on firms' debt structure. For example, several studies have recently explored the choice between single and multiple borrowing in environments with scarce information. Studying the implications of information sharing arrangements for this choice could yield new insights into the role of multiple borrowing and firms' debt structure.

Appendix

Variable definitions.

This table reports definitions of the variables used in the empirical analysis.

Variable	Variable definition
<i>Measures of moderate delinquency</i>	
Average DPD (days)	Average over the life of the contract of the number of days past due

(footnote continued)

interpretation, the estimated reduction in the size of the deal for low rating firms would mainly reflect a drop in the interest rate charged.

Maximum DPD (days)	Maximum over the life of the contract of the number of days past due
Average DPD	$\log(1 + \text{average over the life of the contract of the number of days past due})$
Maximum DPD	$\log(1 + \text{maximum over the life of the contract of the number of days past due})$
No. of times over 30 DPD	$\log(1 + \text{number of times in which days past due are 30 or more})$
Delinquency	Dummy equal to one if Maximum DPD is positive, zero otherwise
Current DPD	$\log(1 + \text{number of days past due as of the current quarter})$
<i>Measures of major events of delinquency or default</i>	
No. of times over 90 DPD	$\log(1 + \text{number of times in which days past due are 90 or more})$
Serious delinquency	Dummy equal to one if No. of times over 90 DPD is positive, zero otherwise
Major event of default	Dummy equal to one if there is a major event of default (e.g., a legal action, debt collection, forced contract extension), zero otherwise
Bankruptcy	Dummy equal to one if there is bankruptcy, zero otherwise
<i>Other variables</i>	
Information	In baseline specifications, dummy equal to one if the contract start date is after the lender's entry into the bureau, zero otherwise (for alternative definitions, see the single tables)
Loan dummy	Dummy equal to one if the contract is a loan (or related contract), zero otherwise
Contract maturity	$\log(\text{scheduled contract maturity, in months})$
Contract size	$\log(\text{total amount to be repaid by the firm over the contract life})$
Guarantor	Dummy equal to one if the contract has a guarantor (personal and/or corporate), zero otherwise
Firm age	Months since the inception of the firm
Firm rating	Credit rating assigned by PayNet to the firm (as of the start date of the contract)

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